

Climate Project Thursday

July 19, 2024

1 Project: Who's to blame for climate change?

In 1896, Swedish scientist Svante Arrhenius first predicted that changes in atmospheric carbon dioxide (CO₂) levels could substantially alter the earth's temperature. Today, we're now keenly aware of how our emissions alter our climate. A continual Climate Clock, on display in New York City and elsewhere, provides an estimate of how long we have to significantly reduce our emissions, in order to avoid the crucial junction of 1.5 degrees (C) of warming.

Every year, countries around the world gather at the UN's COP meetings to discuss how they, collectively, can reduce carbon emissions. Most countries set an emissions target. However, these targets are widely debated: Should countries that manufacture most of the world's goods be allowed to have higher emissions targets, since their manufacturing is crucial to multiple economies worldwide? Should countries with more urban density be expected to have lower emissions targets, since their citizens tend to travel less? Should countries with extremely cold or warm weather be allowed more emissions for cooling or heating their homes?

In this project, we investigate the factors that are most strongly predictive of a country's emissions. Then, we use our models to predict which countries will be the highest and lowest polluters. In doing so, our models can provide a more informed view of the amount of emissions that can be expected by each country in future years - along with recommended ways to reduce those emissions.

The Dataset - co2_dataset Description This dataset is a comprehensive collection of socio-economic and environmental indicators for various countries over multiple years.

Variables - **country**: name of country - **year**: year the data was recorded - **area**: countries total area in square kilometers - **agriculture_land_per**: percentage of the total land area used for agriculture - **population**: total population of the country - **gdp**: Gross Domestic Product (GDP) measured in current US dollars. GDP is a measure for the overall size of a country's economy. - **agriculture_gdp_per**: percentage of GDP contributed by the agriculture, forestry, and fishing sector. - **industry_gdp_per**: percentage of GDP contributed by the industrial sector, including construction - **manufacturing_gdp_per**: percentage of GDP contributed by the manufacturing sector - **service_gdp_per**: percentage of GDP contributed by the services sector - **import_goods_currency**: total value of goods and services imported, measured in current US dollars - **import_goods_gdp_per**: percentage of GDP represented by imports - **export_goods_currency**: total value of goods and services exported, measured in current US dollars - **export_goods_gdp_per**: percentage of GDP represented by exports - **trade_gdp_per**: percentage of GDP represented by the total trade (imports plus exports) - **agriculture_employment_per**: percentage of people working in agriculture related jobs from a countries total employment - **industry_employment_per**: percentage of people working in indus-

try related jobs from a countries total employment - **service_employment_per**: percentage of people working in service related jobs from a countries total employment - **life_expectancy**: average number of years a newborn is expected to live - **urban_pop_per**: urban population percentage - **urban_growth_per**: urban population percentage growth - **rural_pop_per**: rural population percentage - **rural_growth_per**: rural population percentage growth - **co2_emission**: carbon dioxide emission in metric tons **per capita** (per person)

Data Source All the data for this project was gathered using The World Bank and filtered for this project.

Project Authors: Emma He, Benjamin Salceda, Nate Torres-Lopez, Jason Wier

```
[1]: # Download the coursekata package, which contains helpful R commands
library(coursekata)

# Download the Agri-food CO2 Emission dataset
co2_data <- read.csv('https://skewthescrpt.org/s/co2_dataset.csv')

# Display head of dataset
head(co2_data)
```

CourseKata packages

coursekata 0.15.0		
dslabs	0.8.0	Metrics
0.1.4		
Lock5withR	1.2.2	lsr
0.5.2		
fivethirtyeightdata	0.1.0	mosaic
1.9.1		
fivethirtyeight	0.6.2	supernova
3.0.0		

A data.frame: 6 × 24

	country	year	area	agriculture_land_per	population	gdp	agriculture
	<chr>	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	Albania	1996	27400	41.27737	3168033	3199640826	36.4108
2	Albania	1997	27400	41.42336	3148281	2258513974	31.5434
3	Albania	1998	27400	41.56934	3128530	2545964541	28.7856
4	Albania	1999	27400	41.78832	3108778	3212121651	25.9187
5	Albania	2000	27400	41.75182	3089027	3480355258	24.5154
6	Albania	2001	27400	41.56934	3060173	3922100794	22.7161

1.0.1 Project Roadmap

- **1. Exploratory Data Analysis (EDA)** - Look for key predictors and unusual features in our dataset.
- **2. Data Processing & Baseline Model** - Process the data and make a baseline model (the mean model) to predict **co2_emission**.

- **3. Advanced Models, Regression** - Make multiple regression models to predict `co2_emission` from countries' predictor variables.
- **4. Advanced Models, Neural Networks** - Make neural networks to predict `co2_emission` from countries' predictor variables.
- **5. Evaluation & Conclusion**

1.1 Part 1: Exploratory Data Analysis (EDA)

In this section, we will: - Describe the dimensions of our dataset - Visualize key quantitative and categorical variables. Comment on any unusual features. - Visualize the predictor variables in relationship with our target: `co2_emission` (CO2 emissions levels, per person). Identify strong/weak predictors of `co2_emission` and comment on any non-linear trends.

```
[2]: co2_data$co2_total <- co2_data$co2_emission * co2_data$population
head(co2_data)
dim(co2_data)
co2_mean <- mean(co2_data$co2_emission)
co2_mean
```

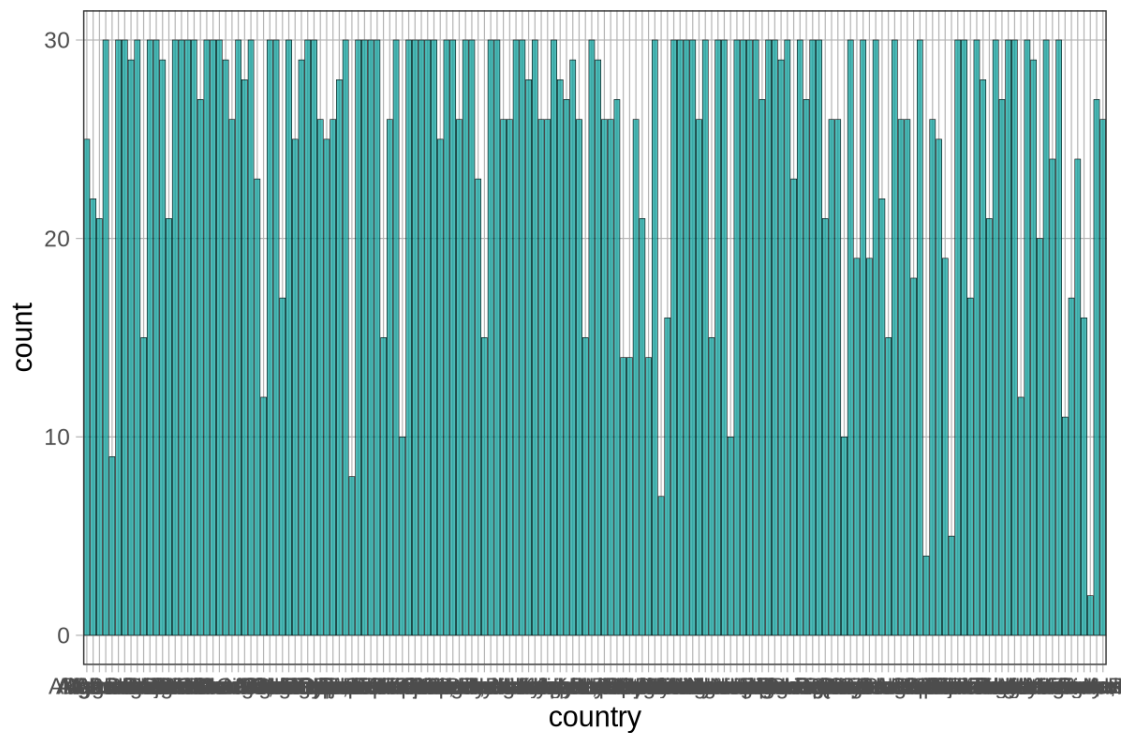
		country	year	area	agriculture_land_per	population	gdp	agriculture
		<chr>	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
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	2	Albania	1997	27400	41.42336	3148281	2258513974	31.5434
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	6	Albania	2001	27400	41.56934	3060173	3922100794	22.7161

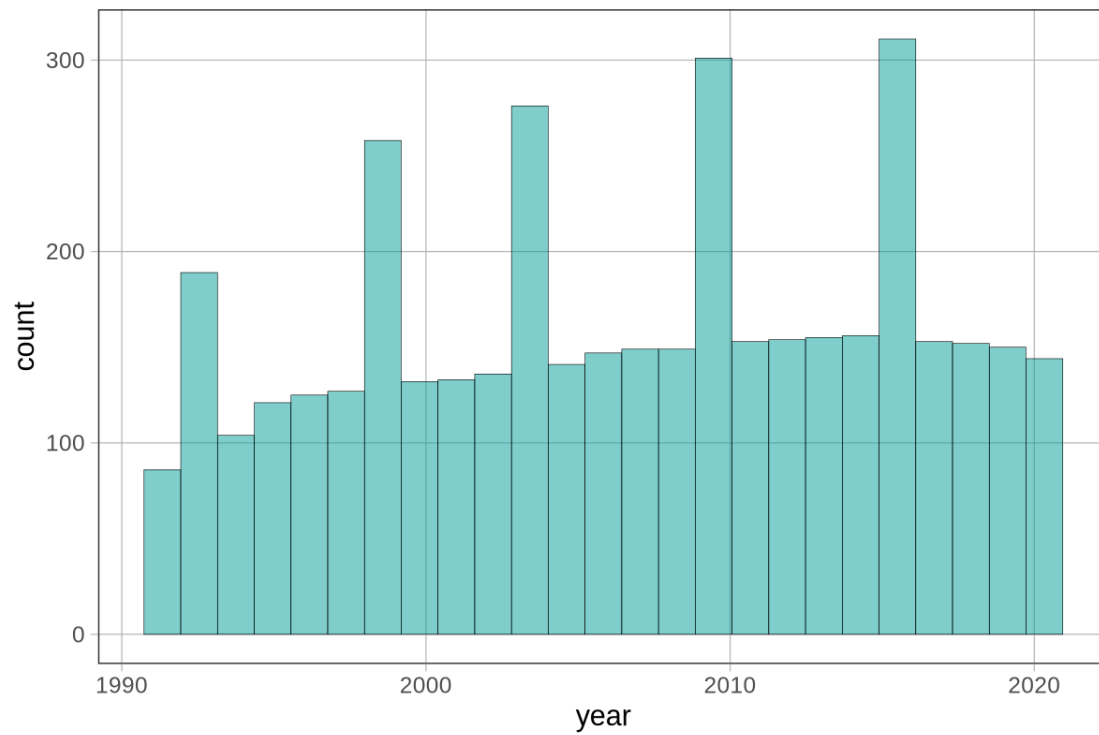
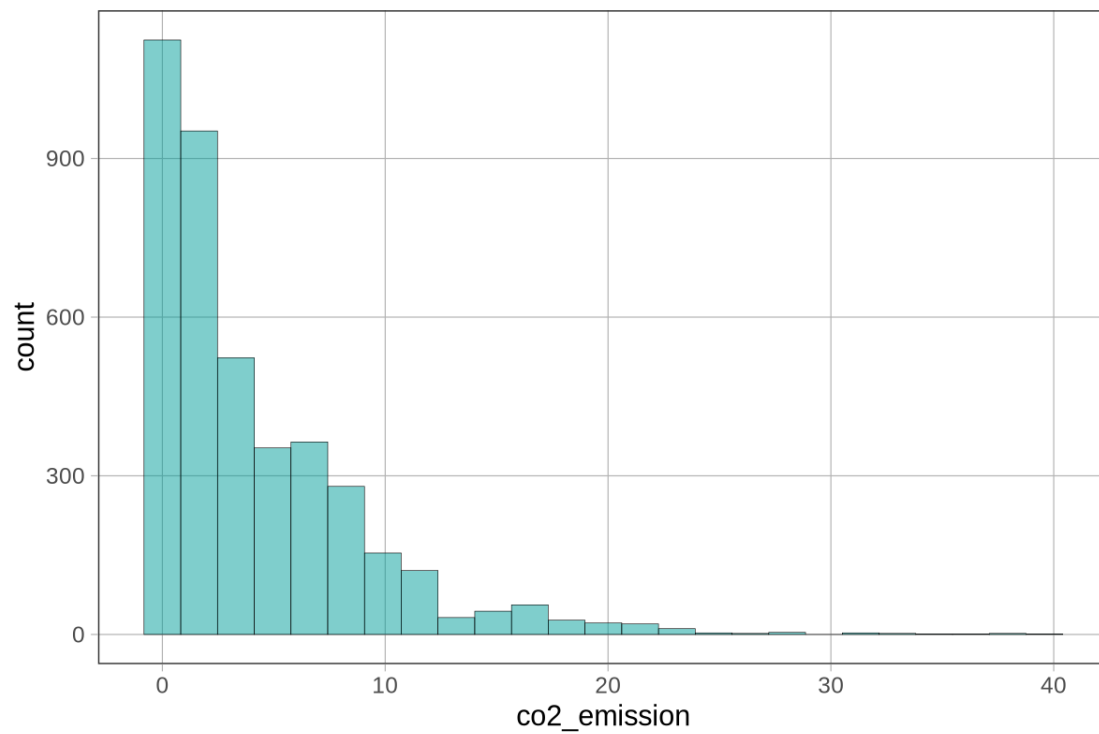
1. 4102 2. 25

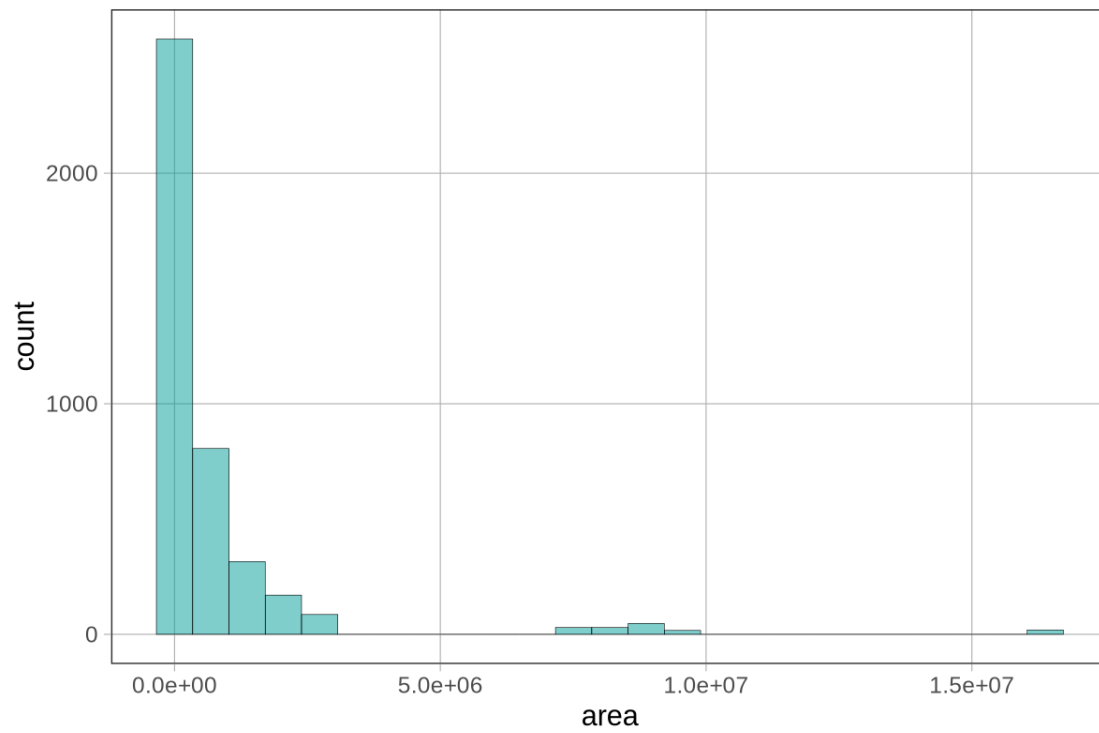
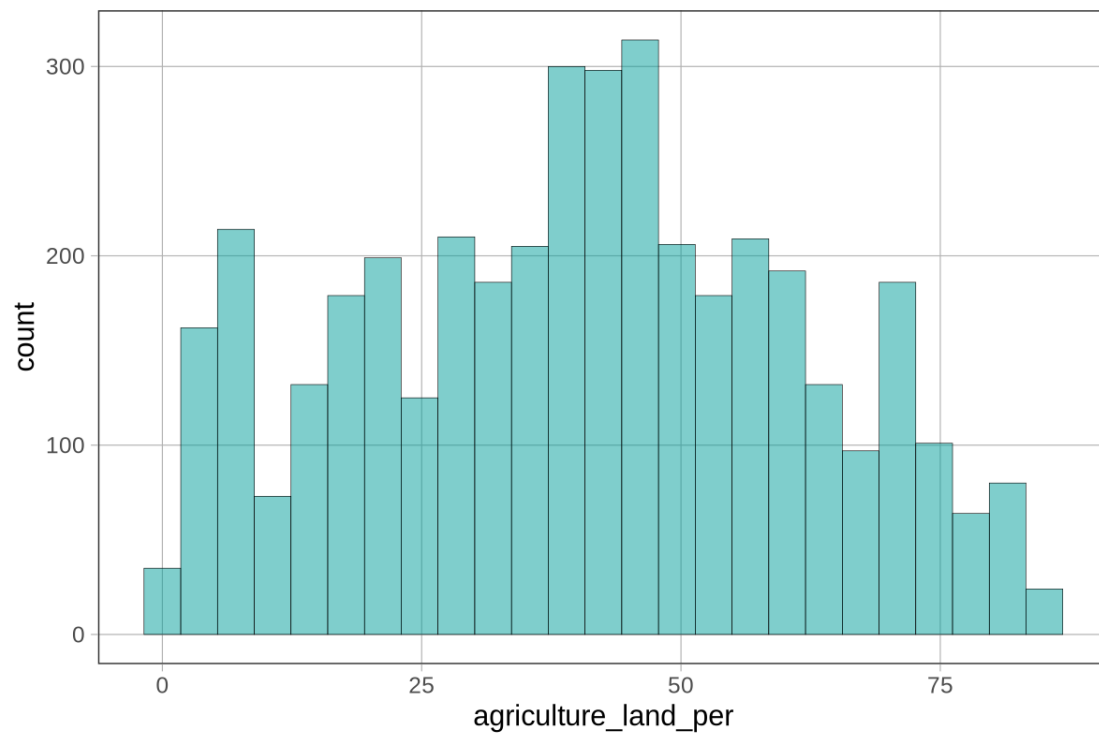
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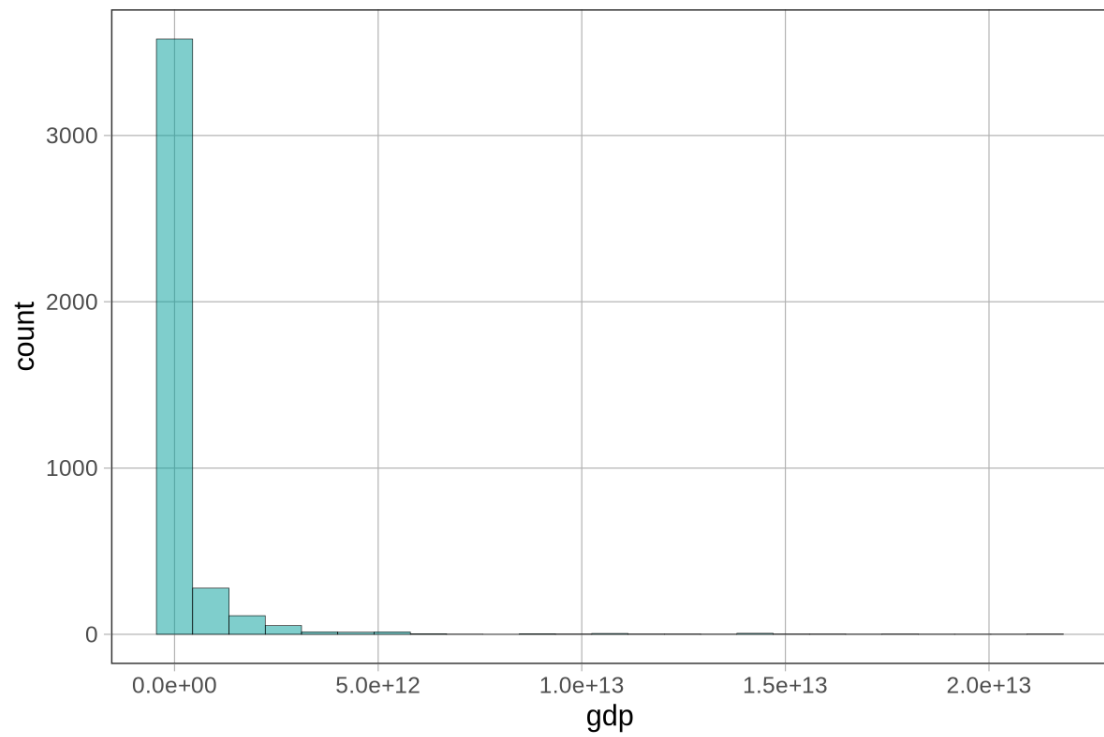
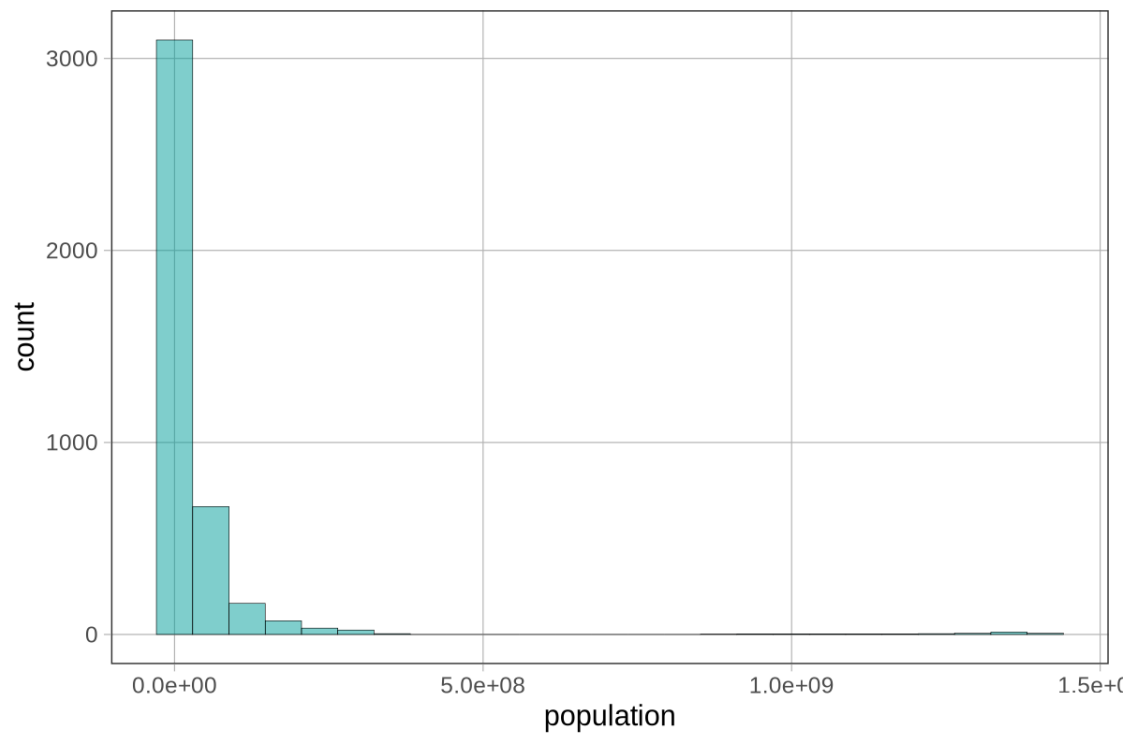
```
[3]: gf_bar(co2_data, ~country)
gf_histogram(~co2_emission, data = co2_data)
gf_histogram(~year, data = co2_data)
gf_histogram(~agriculture_land_per, data = co2_data)
gf_histogram(~area, data = co2_data)
gf_histogram(~population, data = co2_data)
gf_histogram(~gdp, data = co2_data)
gf_histogram(~agriculture_gdp_per, data = co2_data)
gf_histogram(~industry_gdp_per, data = co2_data)
gf_histogram(~manufacturing_gdp_per, data = co2_data)
gf_histogram(~service_gdp_per, data = co2_data)
gf_histogram(~import_goods_currency, data = co2_data)
gf_histogram(~export_goods_currency, data = co2_data)
gf_histogram(~import_goods_gdp_per, data = co2_data)
gf_histogram(~export_goods_gdp_per, data = co2_data)
gf_histogram(~trade_gdp_per, data = co2_data)
gf_histogram(~agriculture_employment_per, data = co2_data)
gf_histogram(~industry_employment_per, data = co2_data)
```

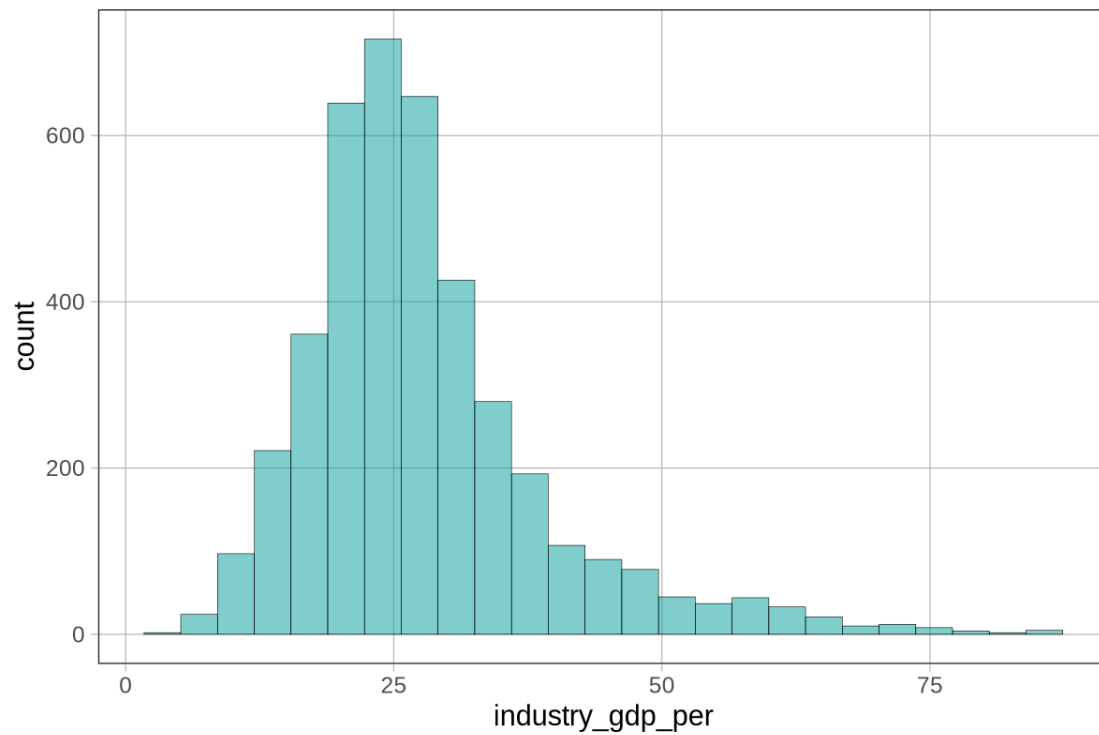
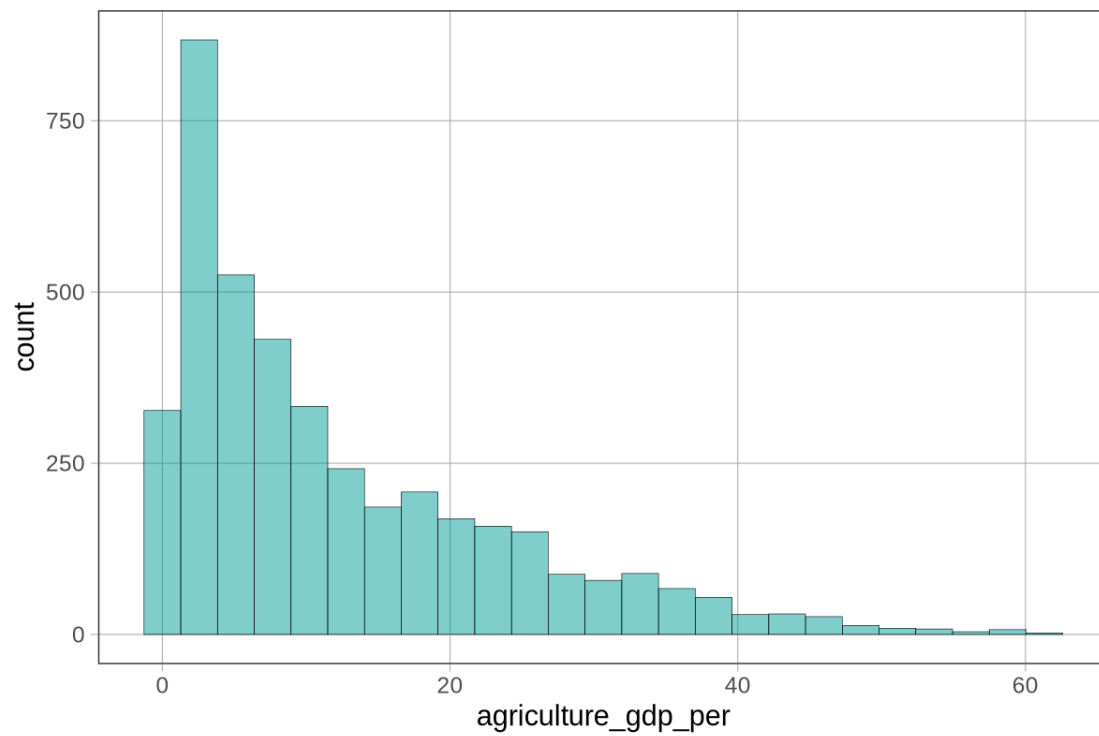
```
gf_histogram(~service_employment_per, data = co2_data)
gf_histogram(~life_expectancy, data = co2_data)
gf_histogram(~urban_pop_per, data = co2_data)
gf_histogram(~urban_growth_per, data = co2_data)
gf_histogram(~rural_pop_per, data = co2_data)
gf_histogram(~rural_growth_per, data = co2_data)
gf_histogram(~co2_total, data = co2_data)
```

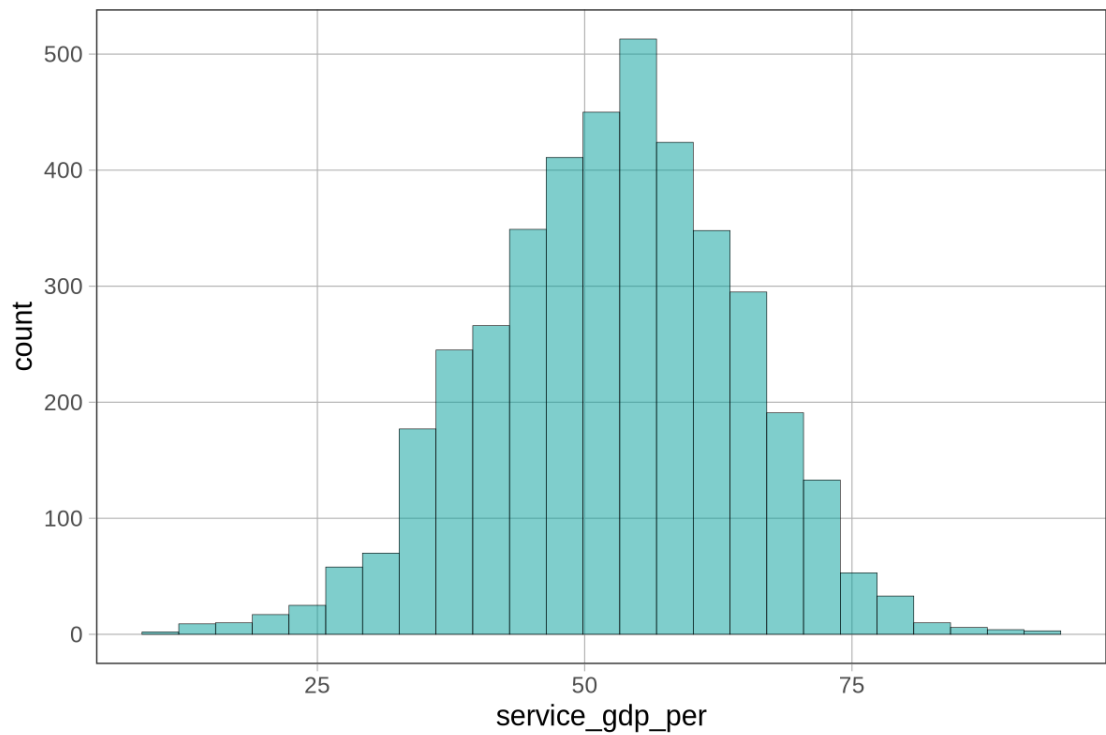
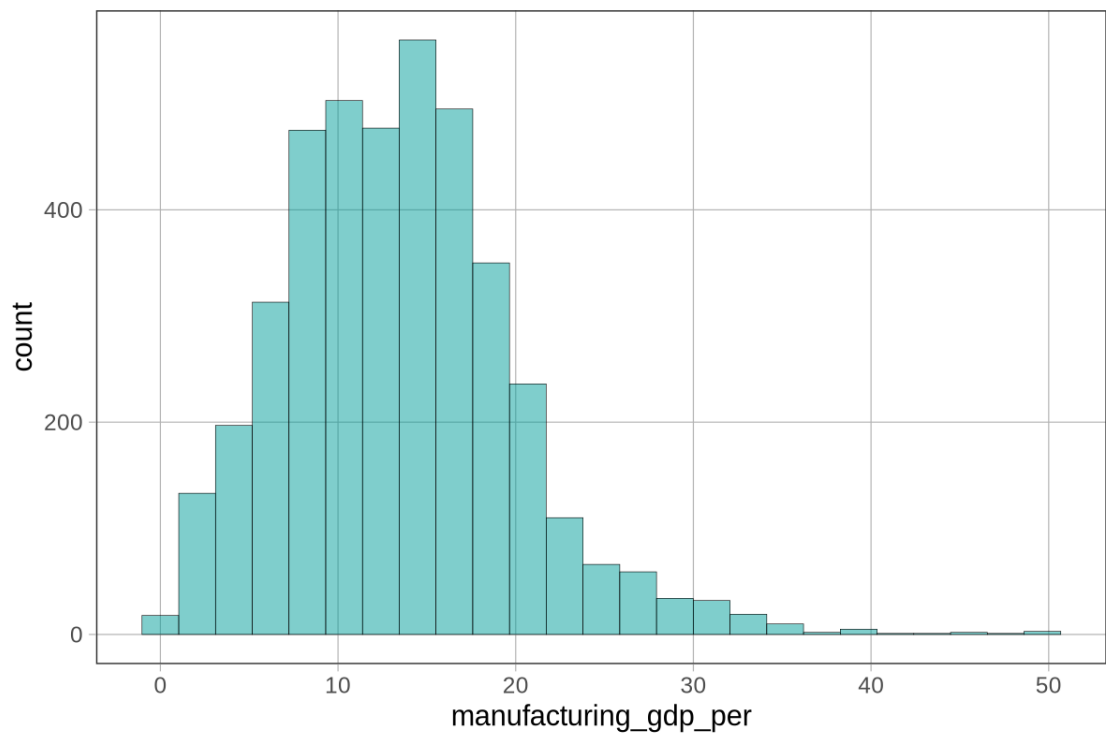


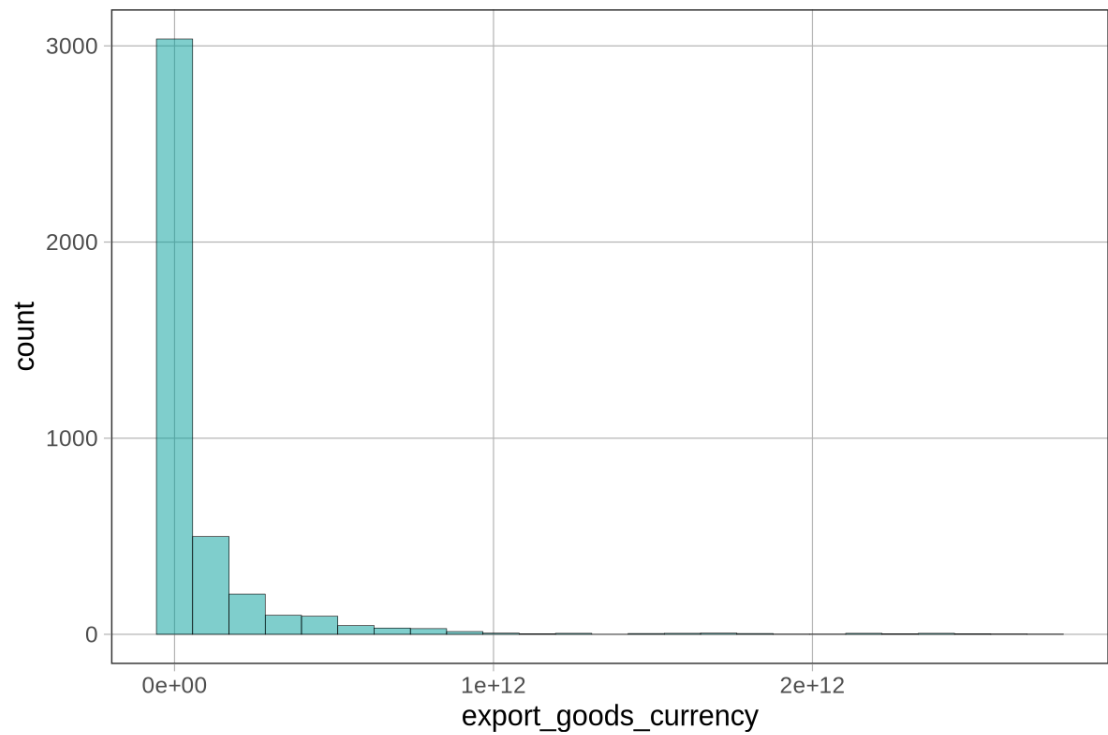
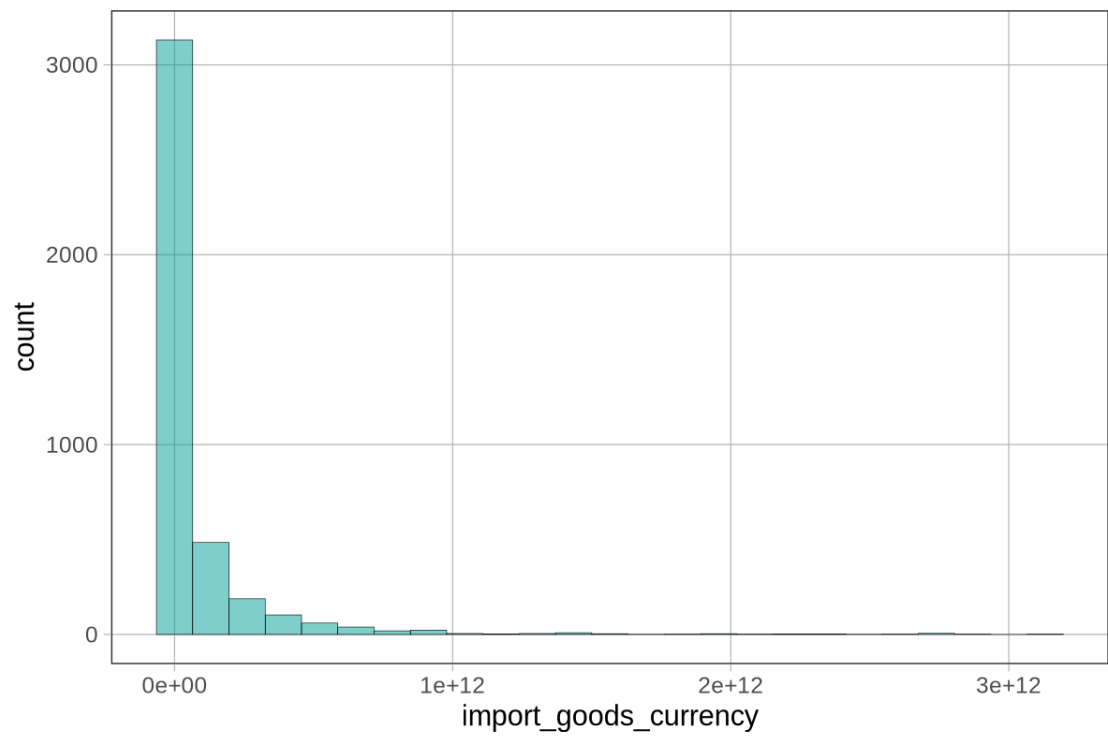


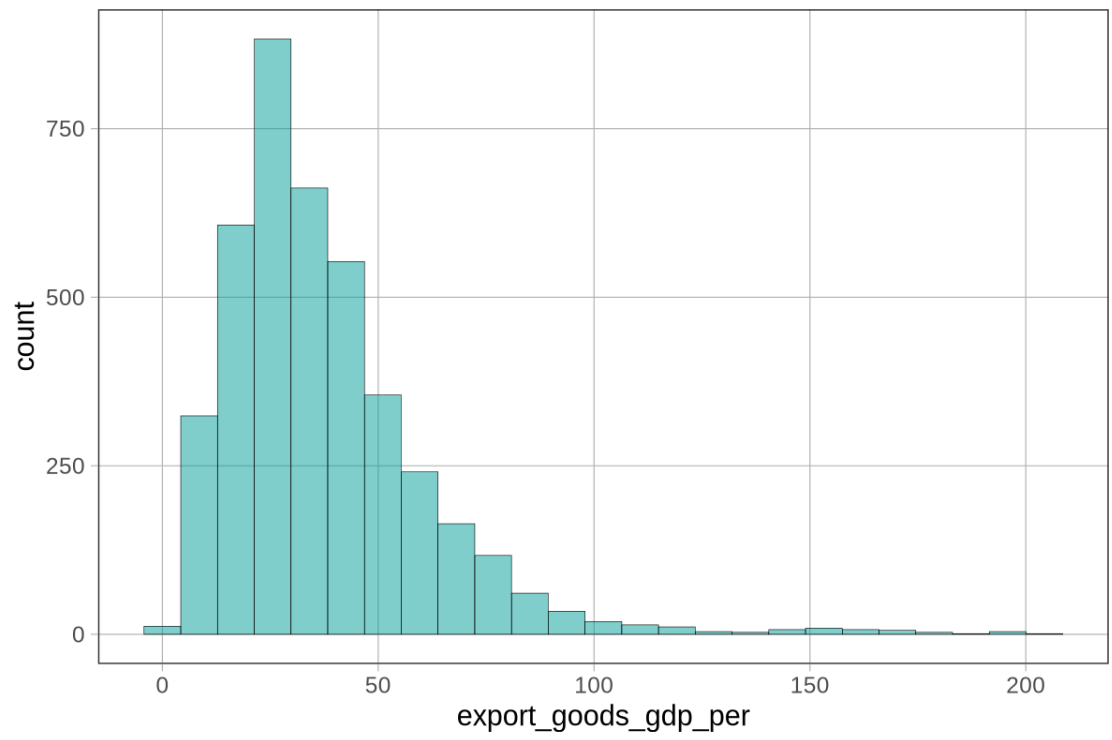
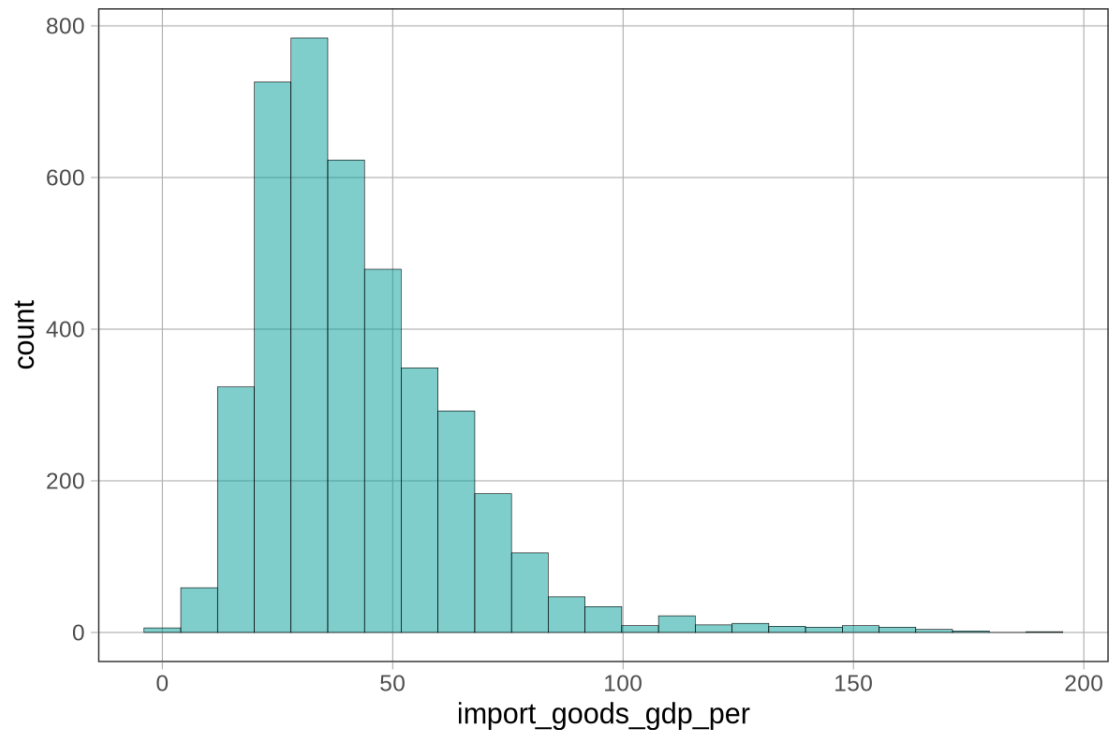


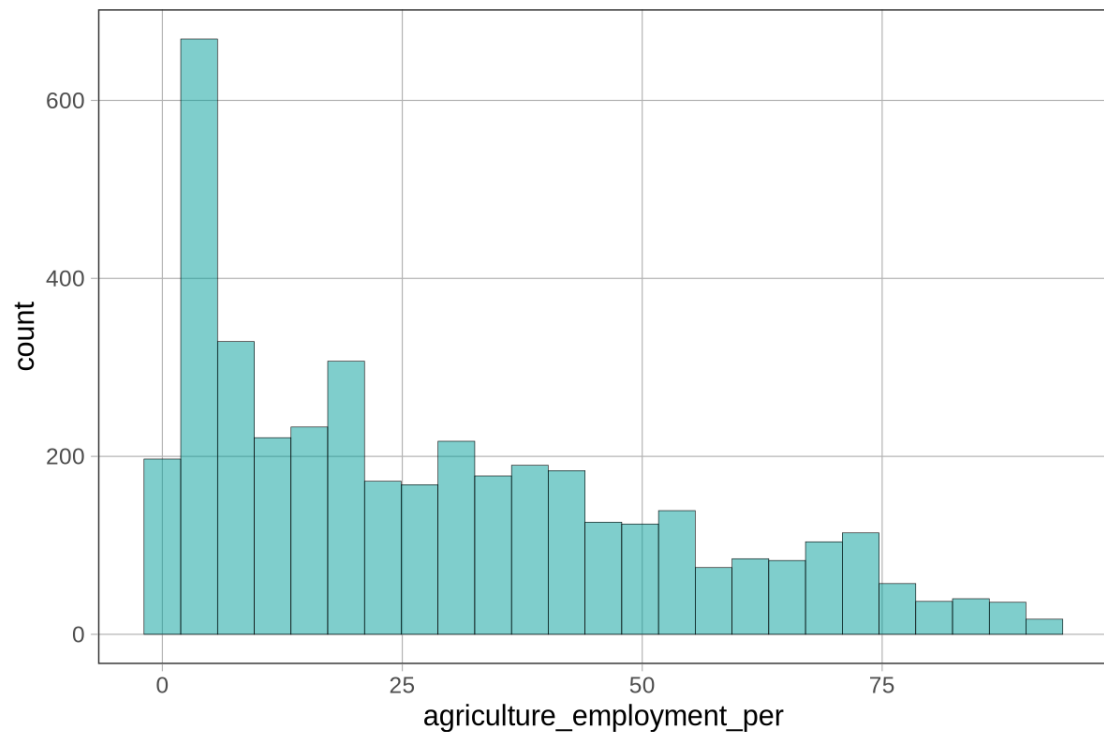
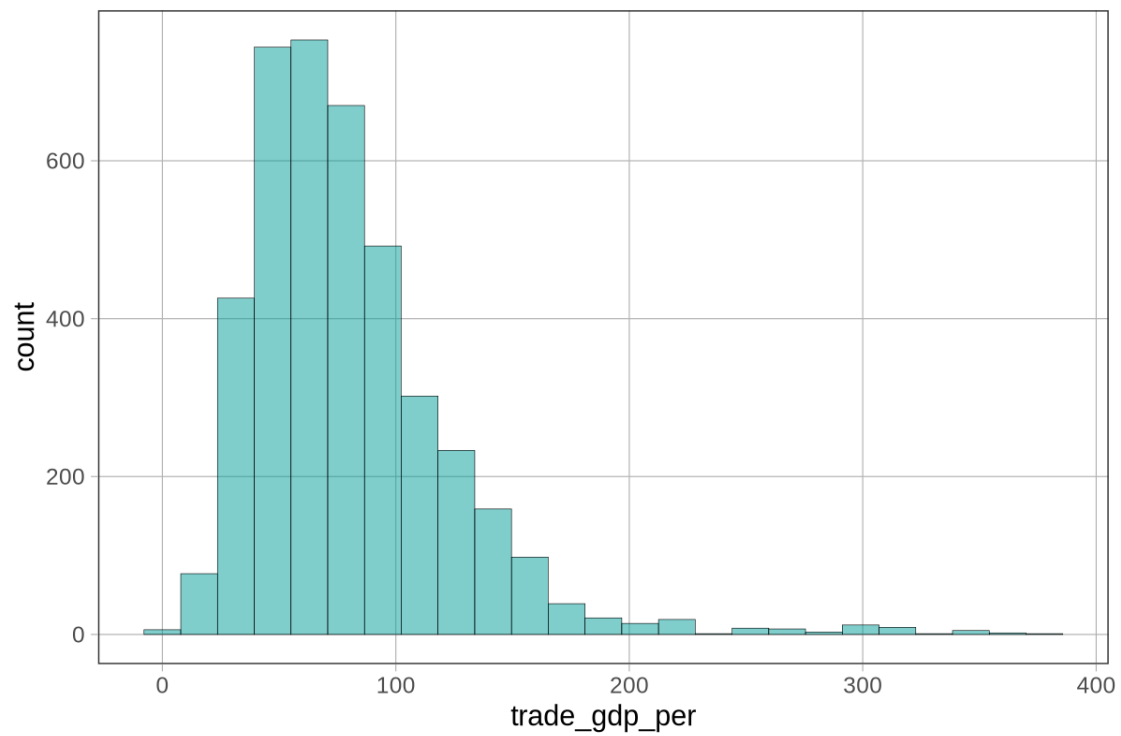


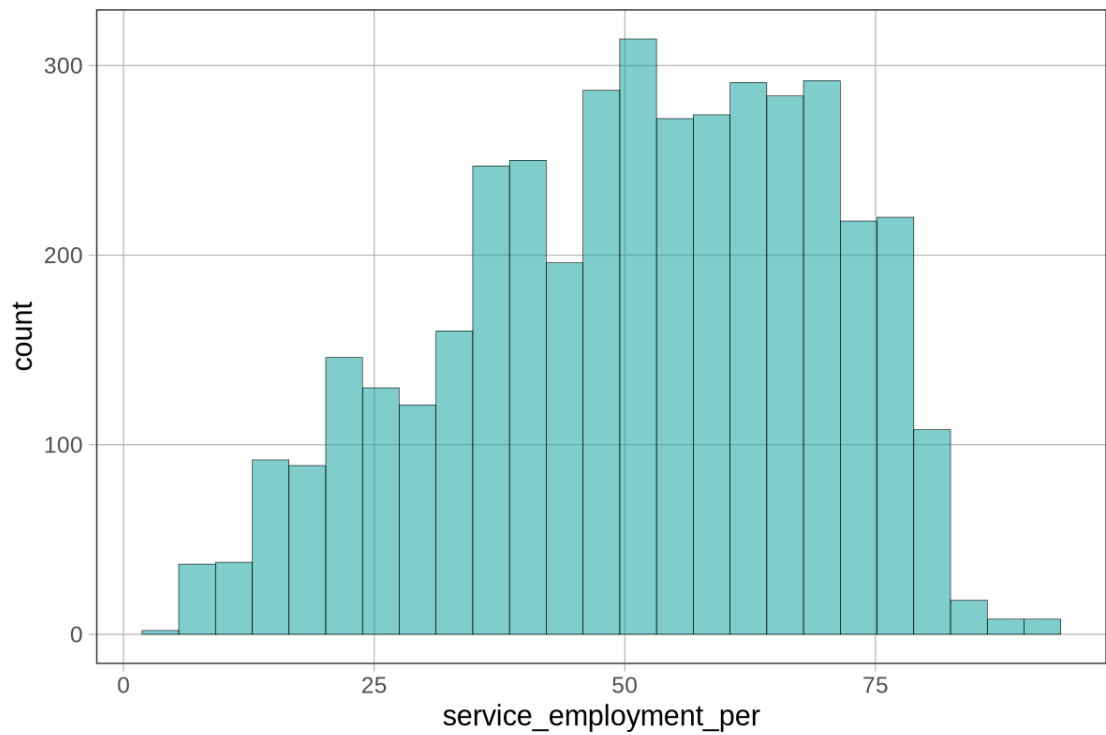
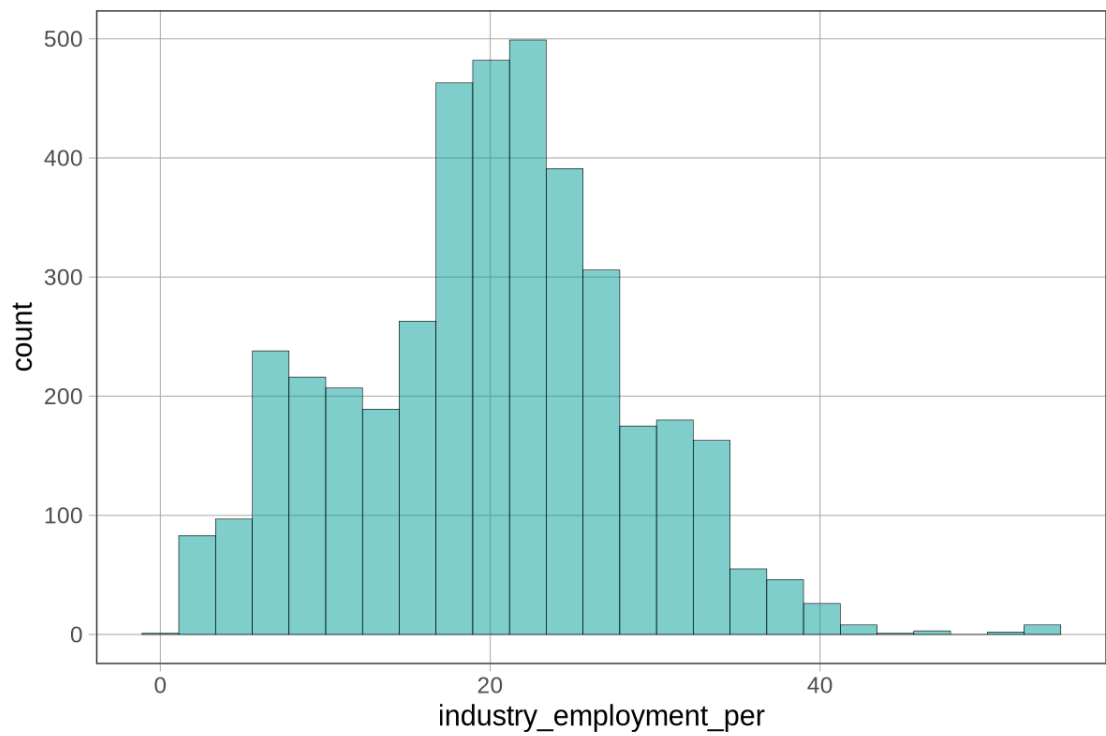


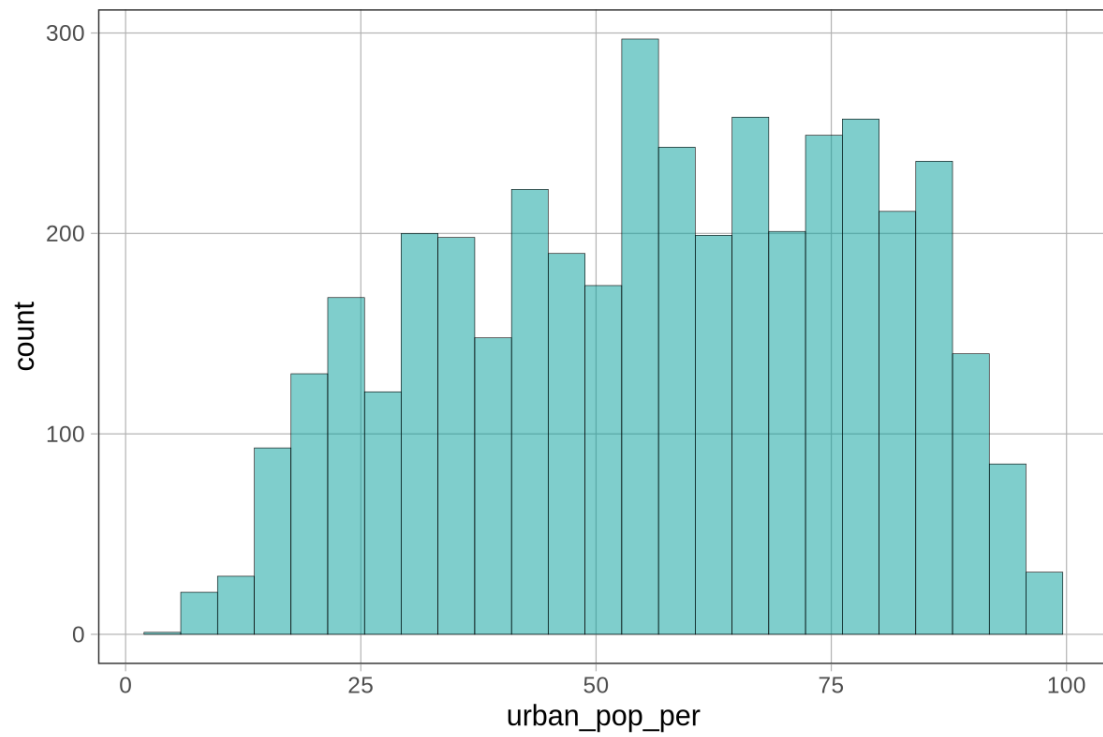
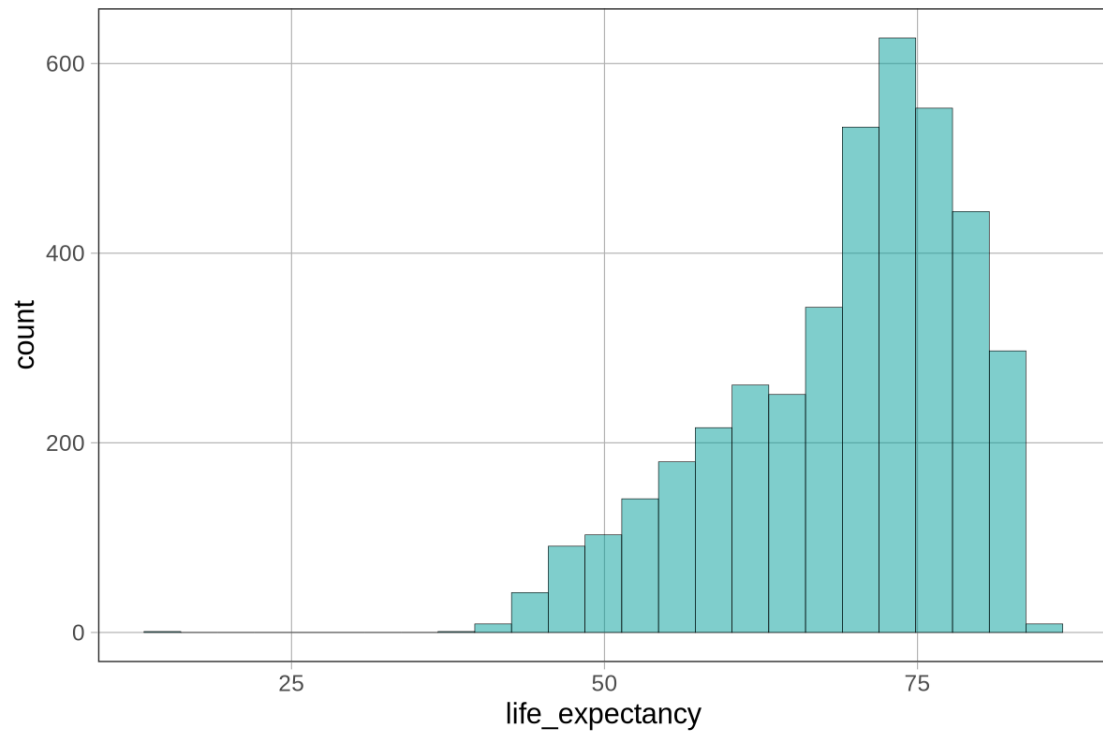


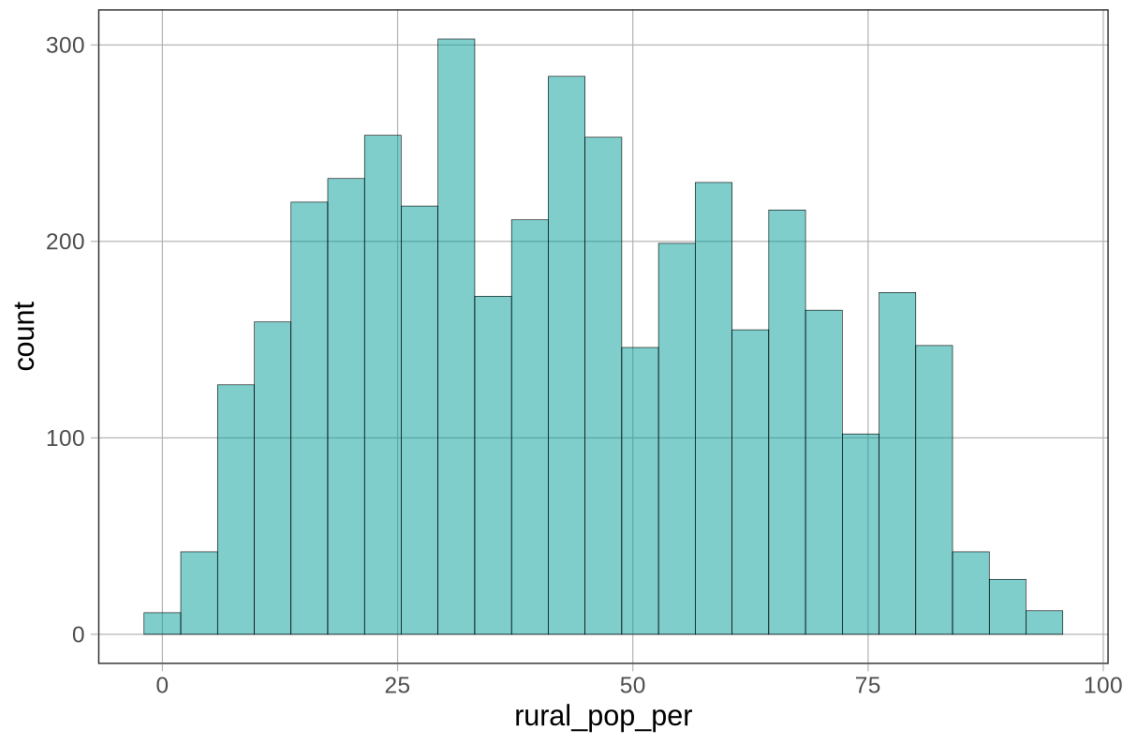
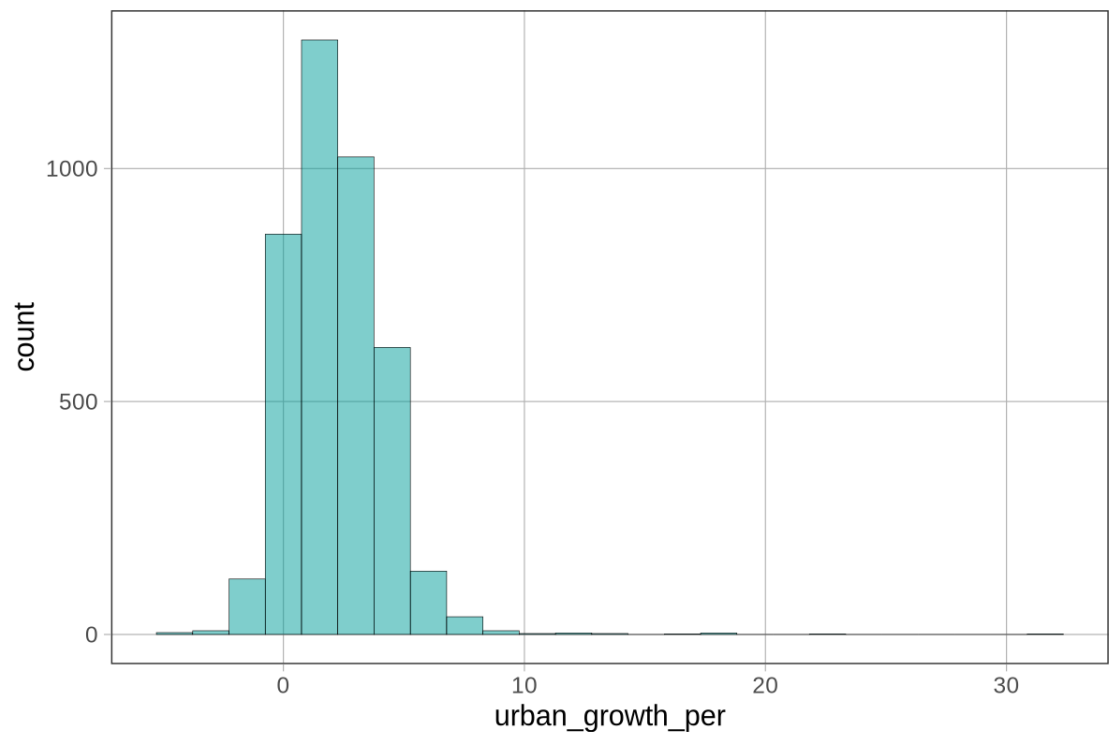


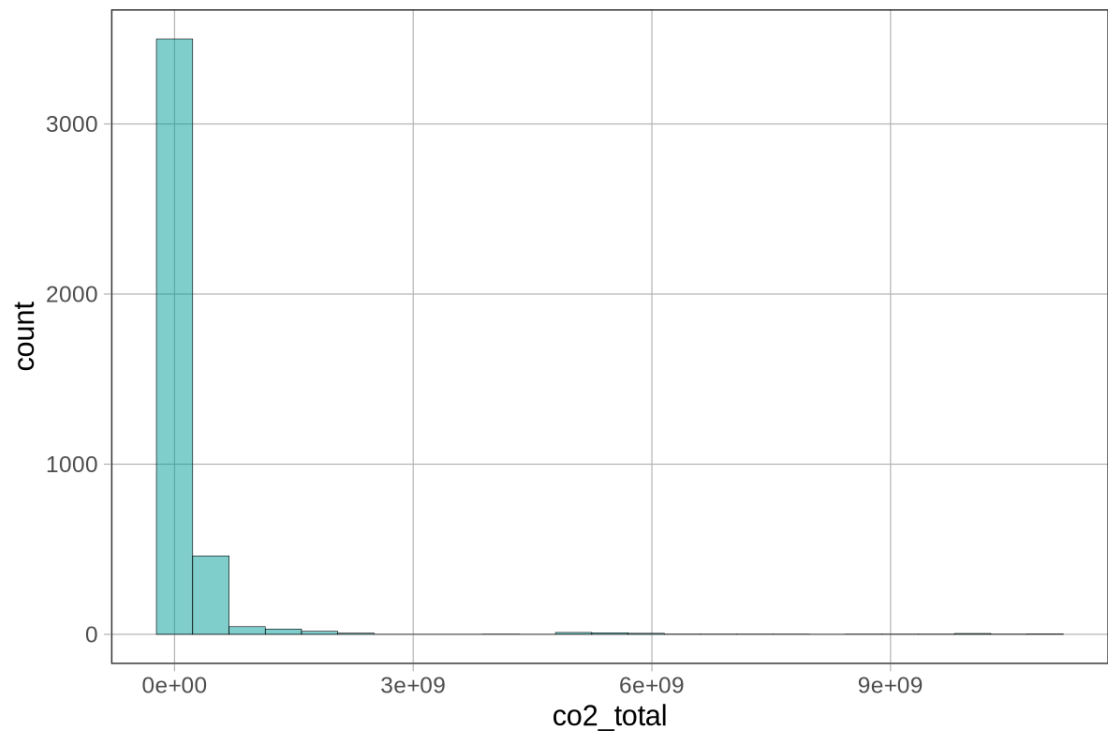
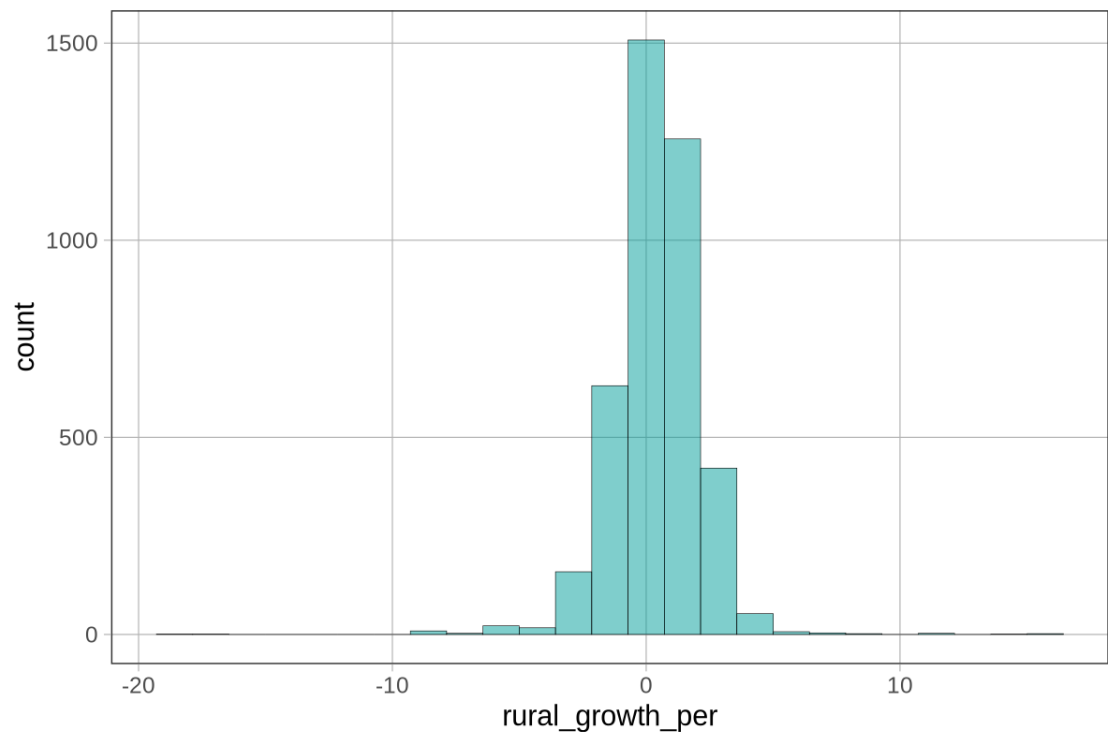




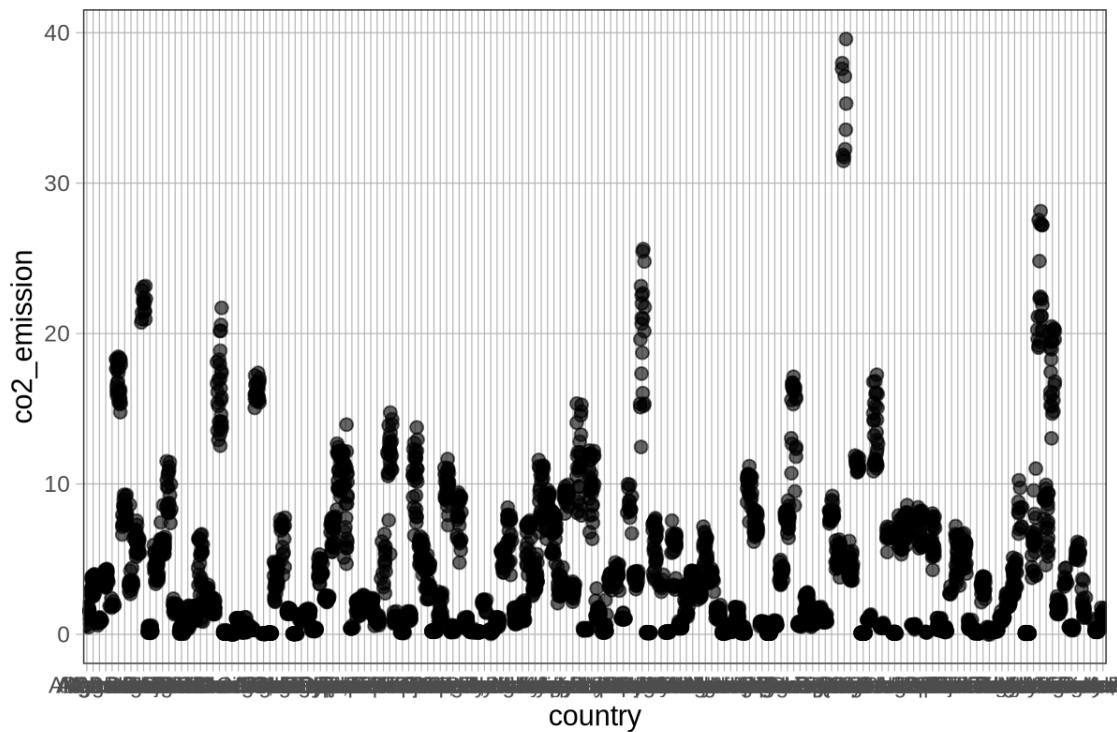


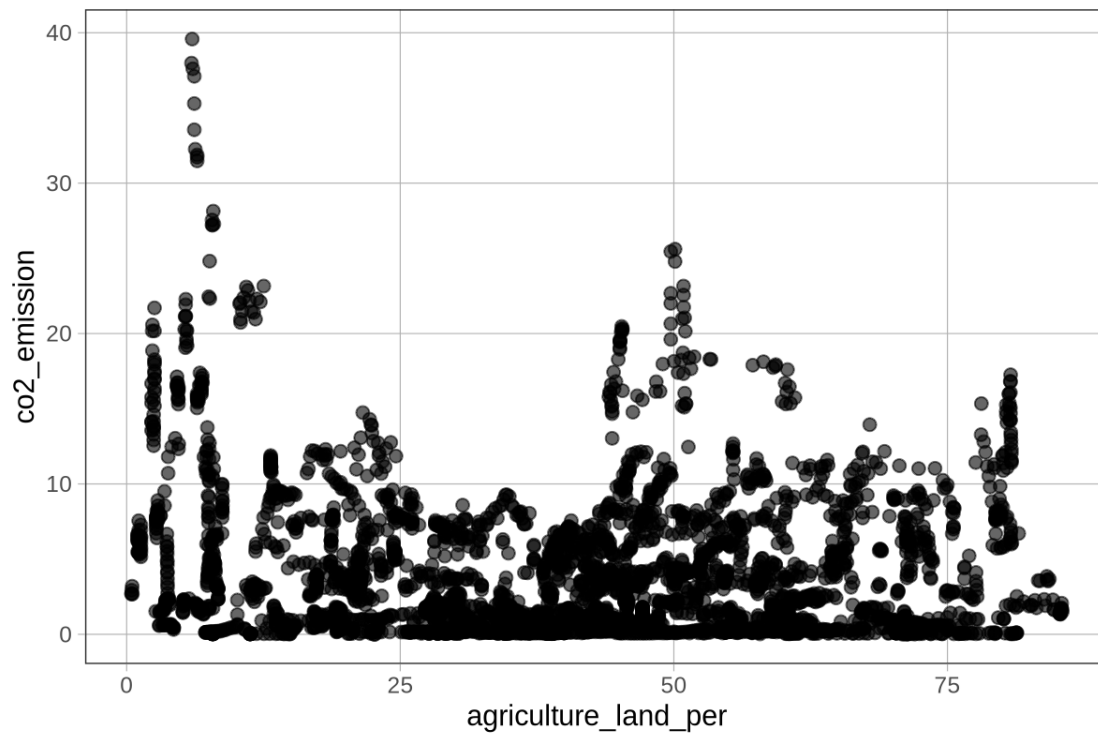
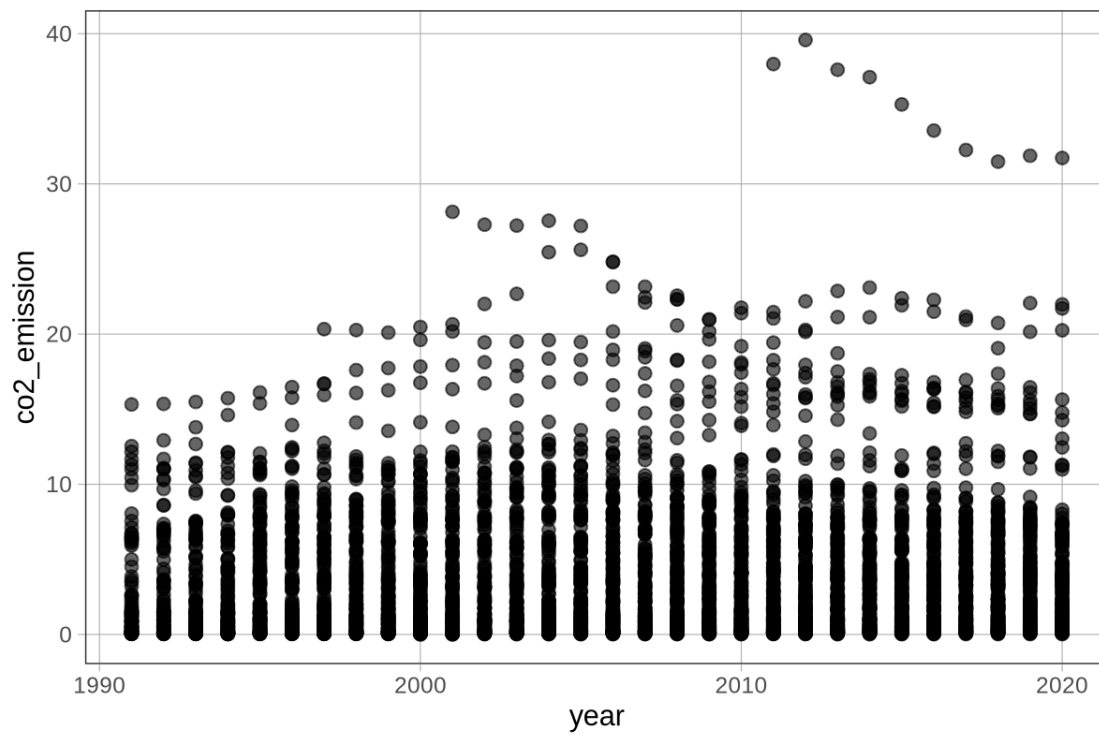


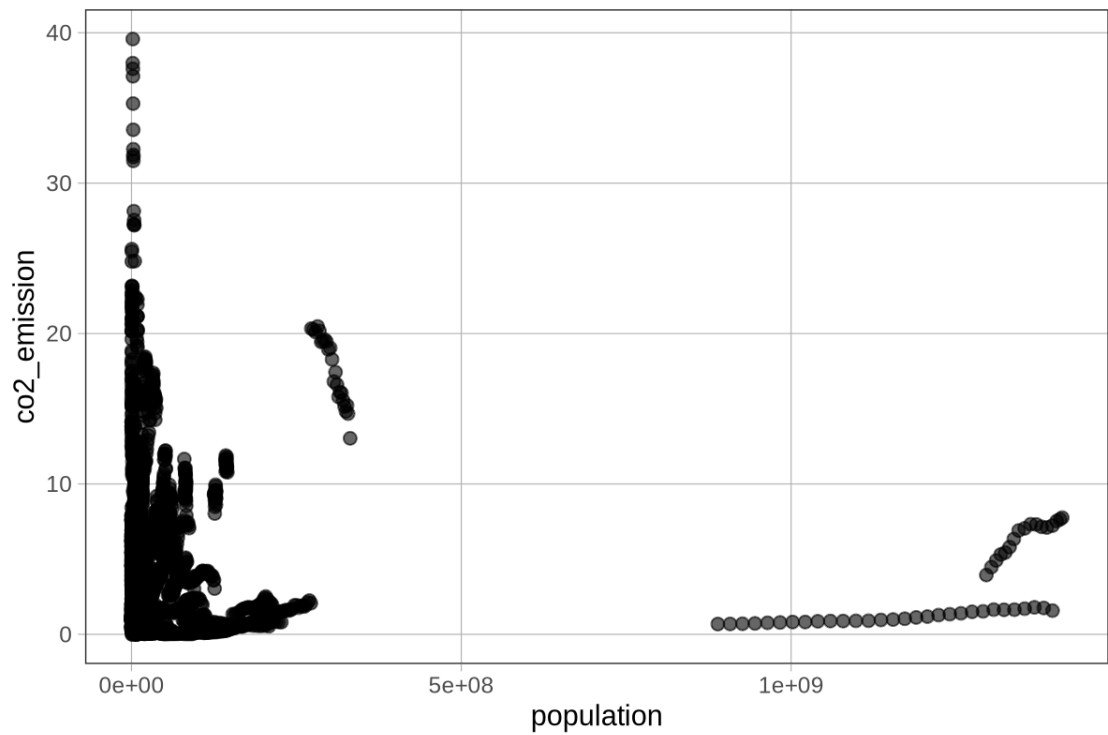
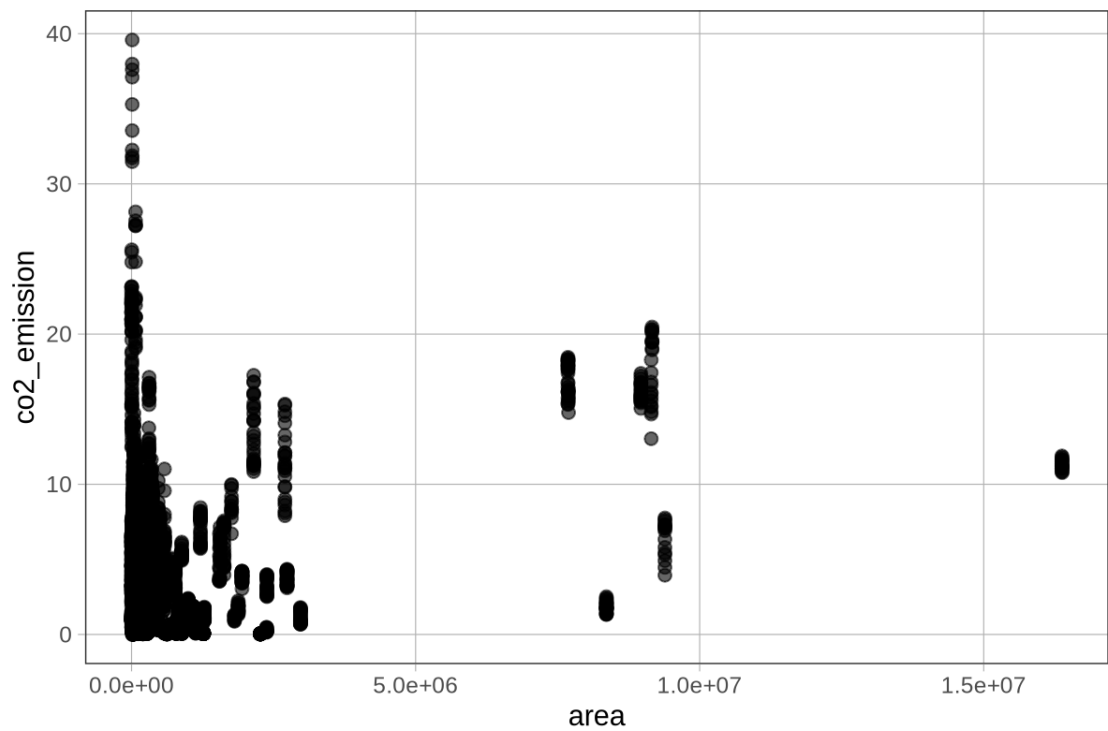


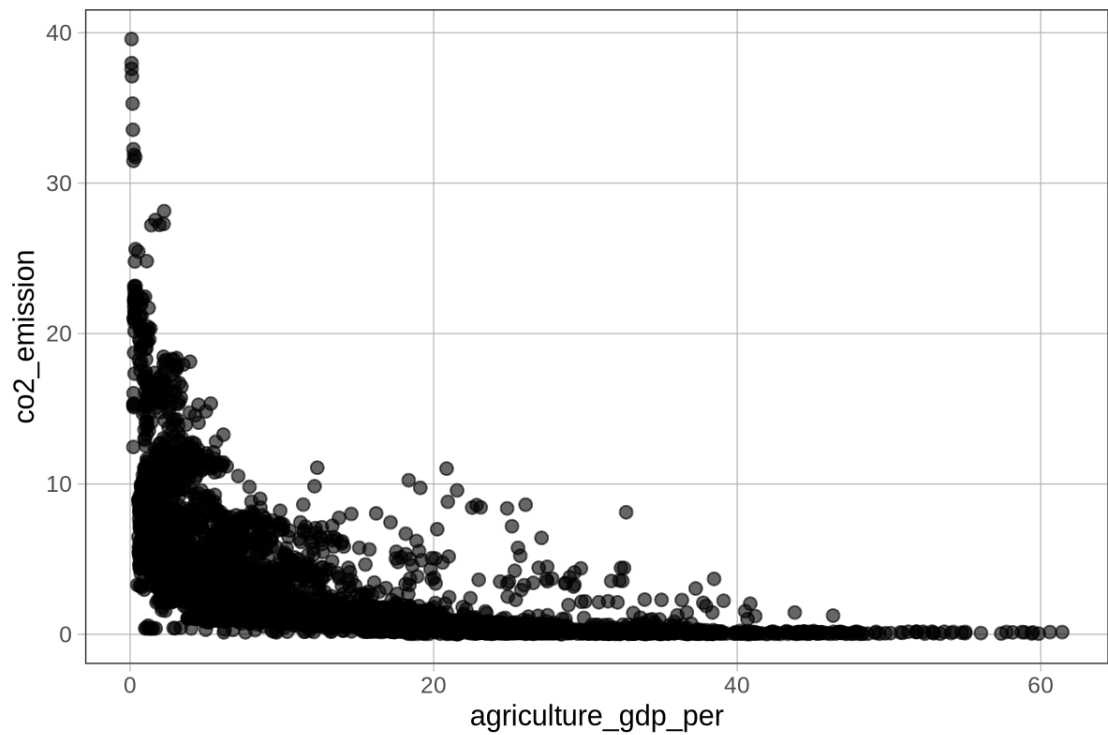
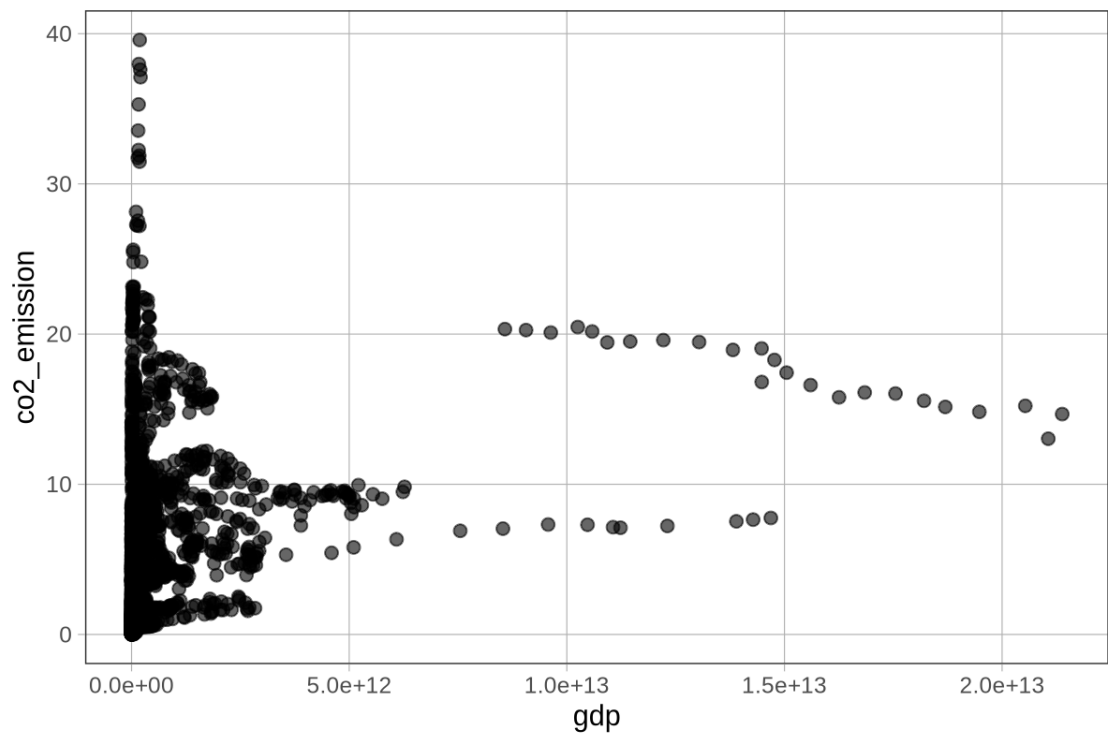


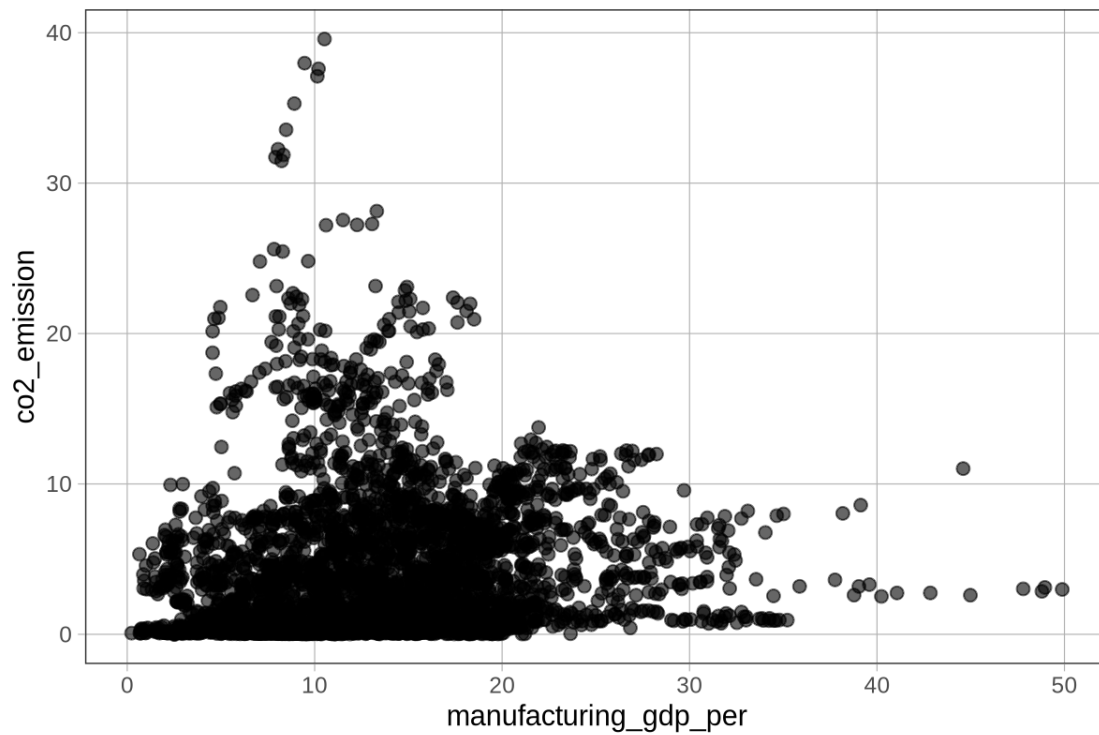
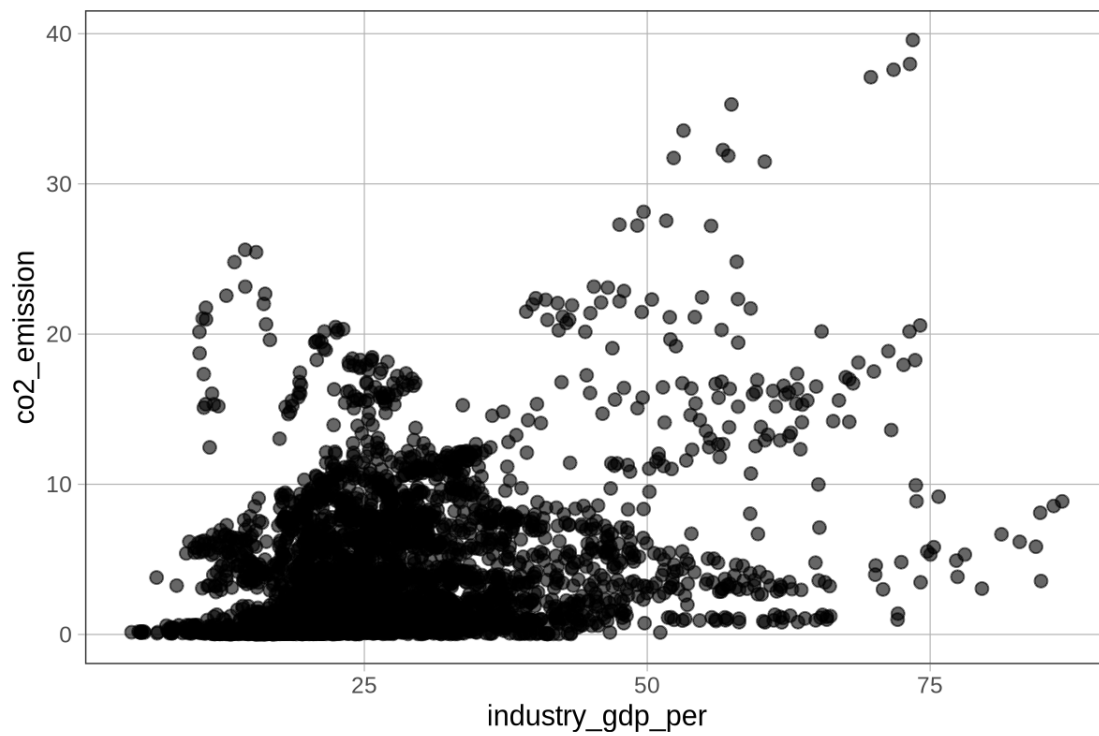
```
[4]: gf_jitter(co2_emission ~ country, data = co2_data)
gf_point(co2_emission ~ year, data = co2_data)
gf_point(co2_emission ~ agriculture_land_per, data = co2_data)
gf_point(co2_emission ~ area, data = co2_data)
gf_point(co2_emission ~ population, data = co2_data)
gf_point(co2_emission ~ gdp, data = co2_data)
gf_point(co2_emission ~ agriculture_gdp_per, data = co2_data)
gf_point(co2_emission ~ industry_gdp_per, data = co2_data)
gf_point(co2_emission ~ manufacturing_gdp_per, data = co2_data)
gf_point(co2_emission ~ service_gdp_per, data = co2_data)
gf_point(co2_emission ~ import_goods_currency, data = co2_data)
gf_point(co2_emission ~ export_goods_currency, data = co2_data)
gf_point(co2_emission ~ import_goods_gdp_per, data = co2_data)
gf_point(co2_emission ~ export_goods_gdp_per, data = co2_data)
gf_point(co2_emission ~ trade_gdp_per, data = co2_data)
gf_point(co2_emission ~ agriculture_employment_per, data = co2_data)
gf_point(co2_emission ~ industry_employment_per, data = co2_data)
gf_point(co2_emission ~ service_employment_per, data = co2_data)
gf_point(co2_emission ~ life_expectancy, data = co2_data)
gf_point(co2_emission ~ urban_pop_per, data = co2_data)
gf_point(co2_emission ~ urban_growth_per, data = co2_data)
gf_point(co2_emission ~ rural_pop_per, data = co2_data)
gf_point(co2_emission ~ rural_growth_per, data = co2_data)
```

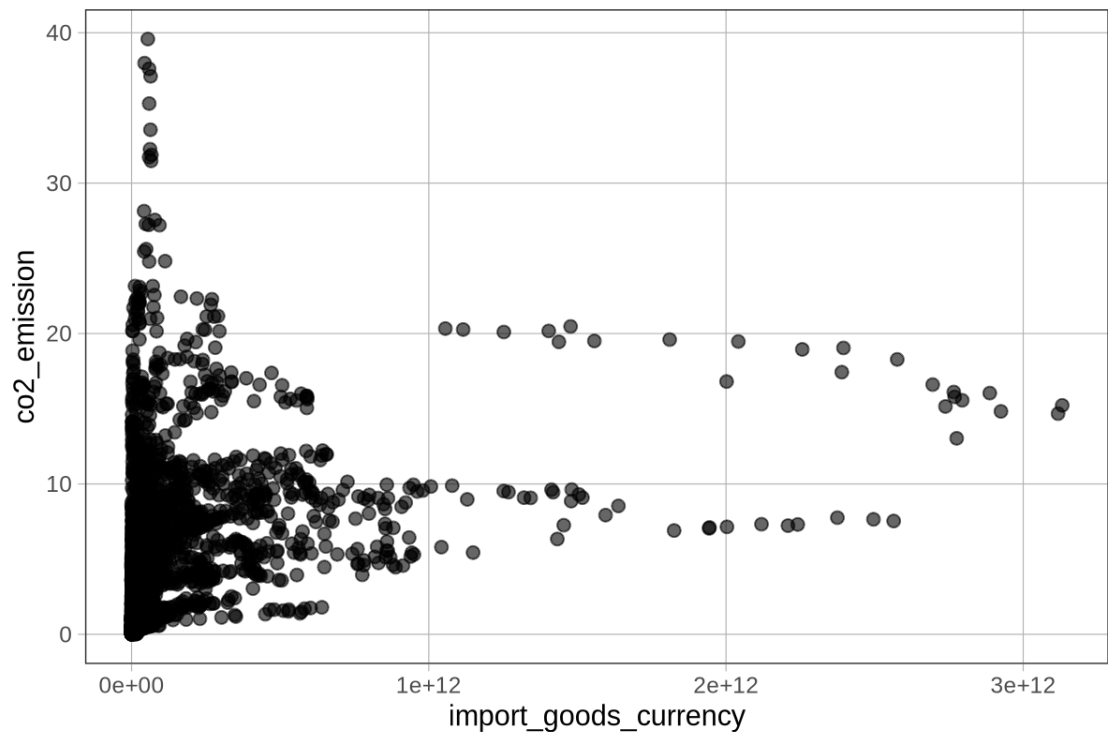
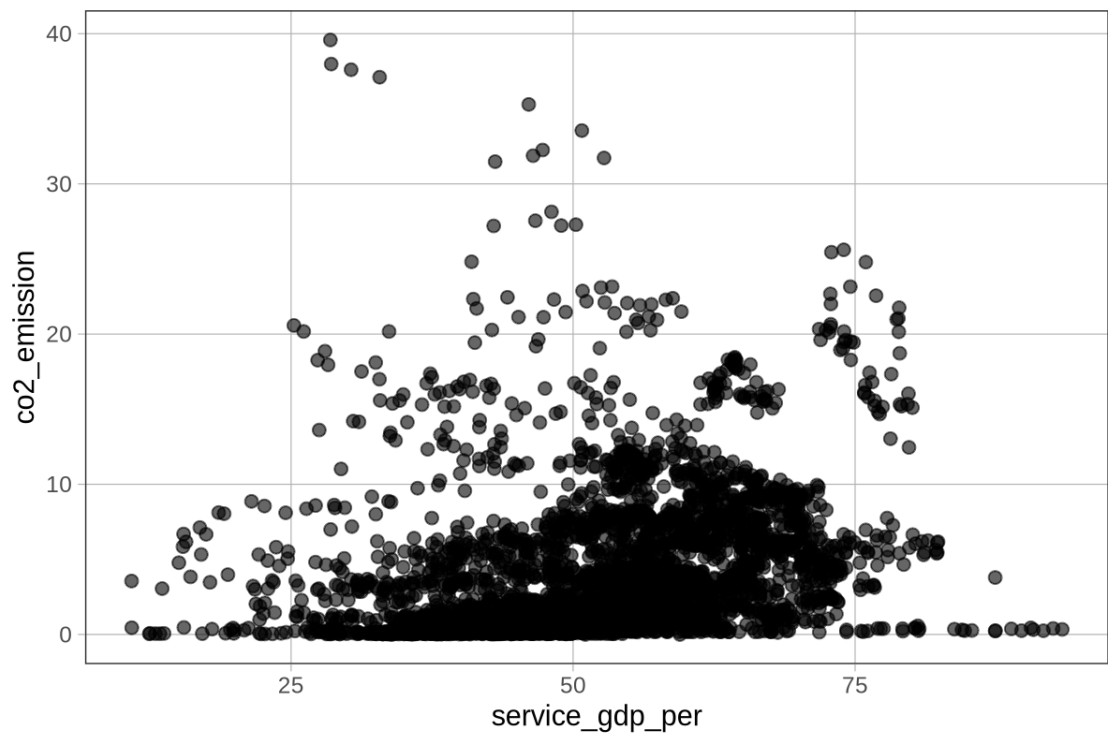


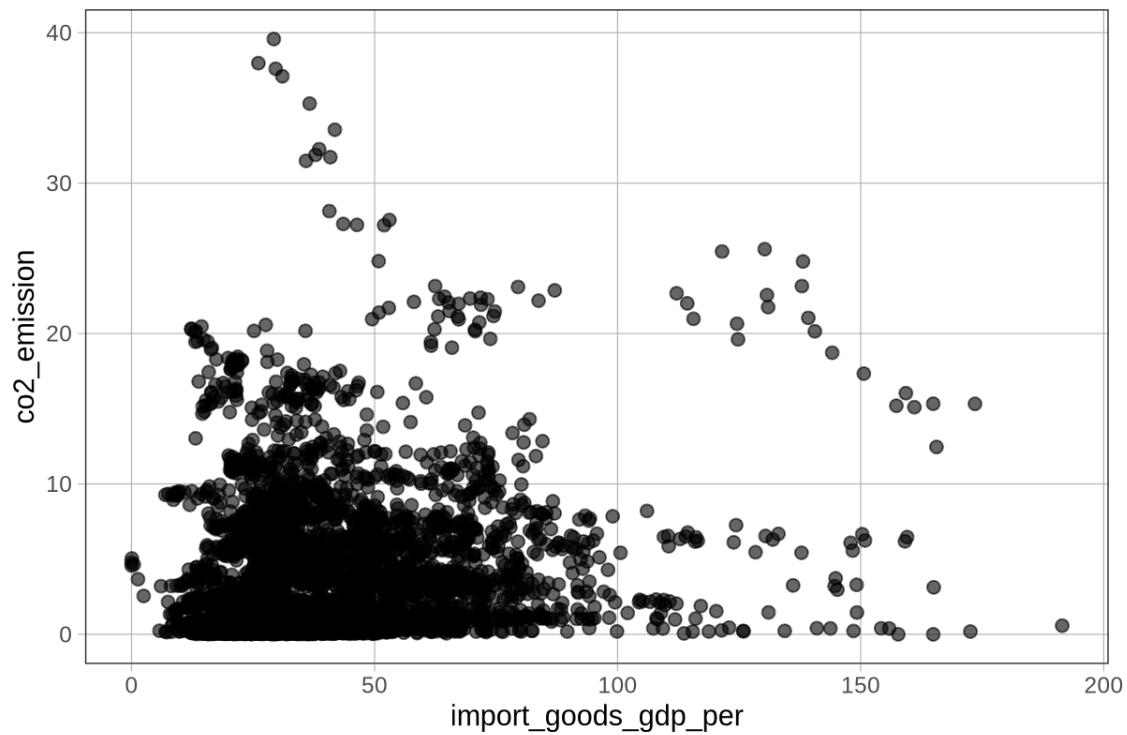
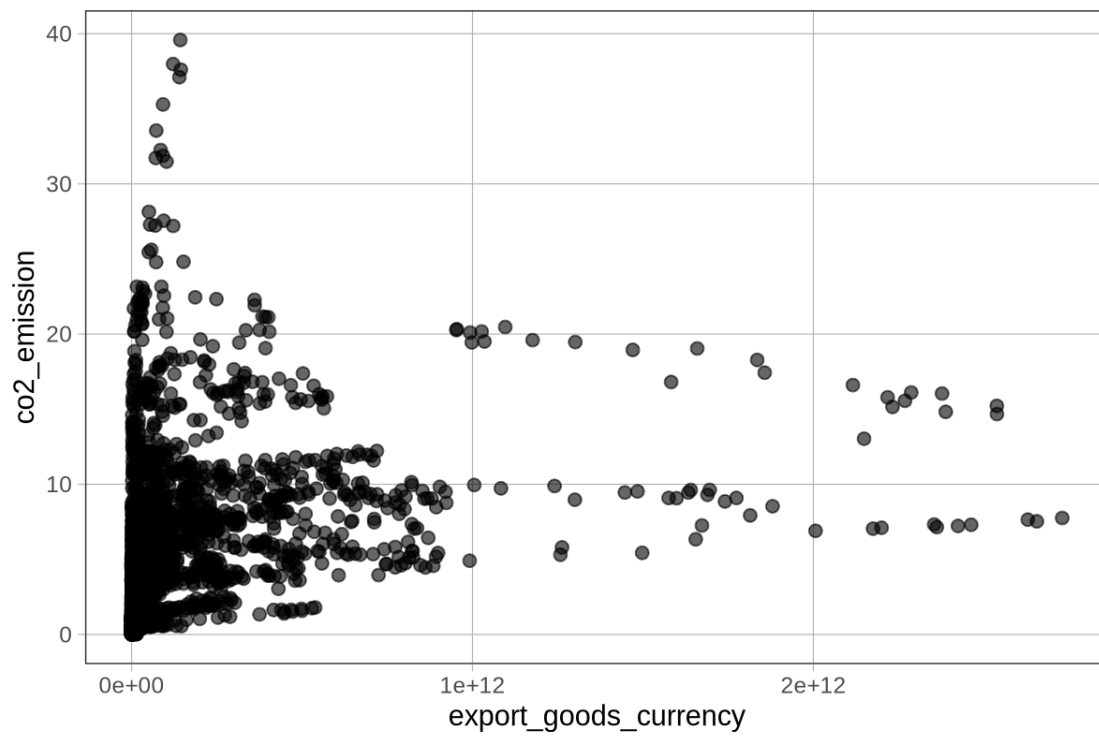


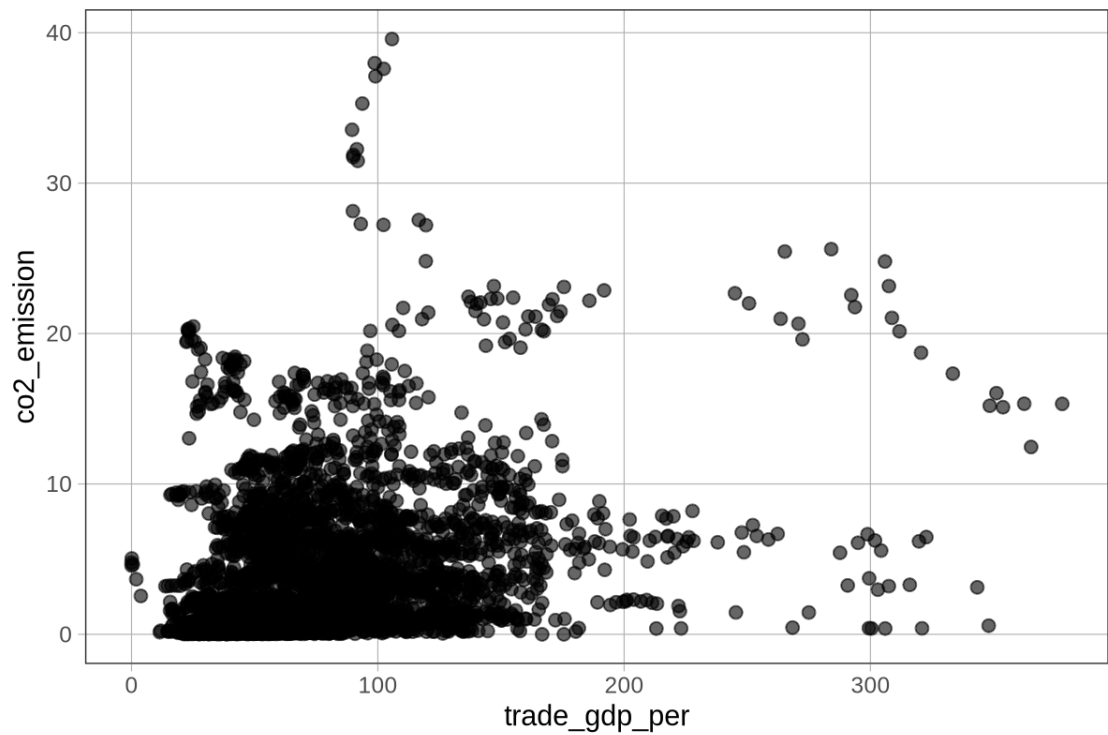
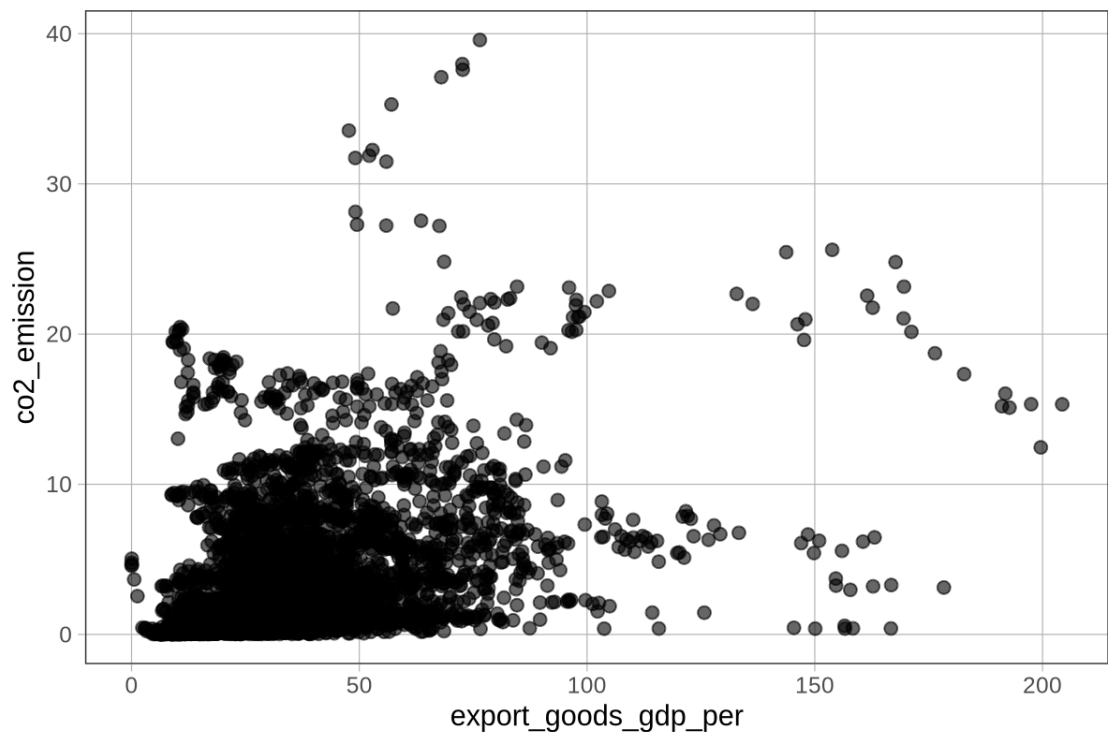


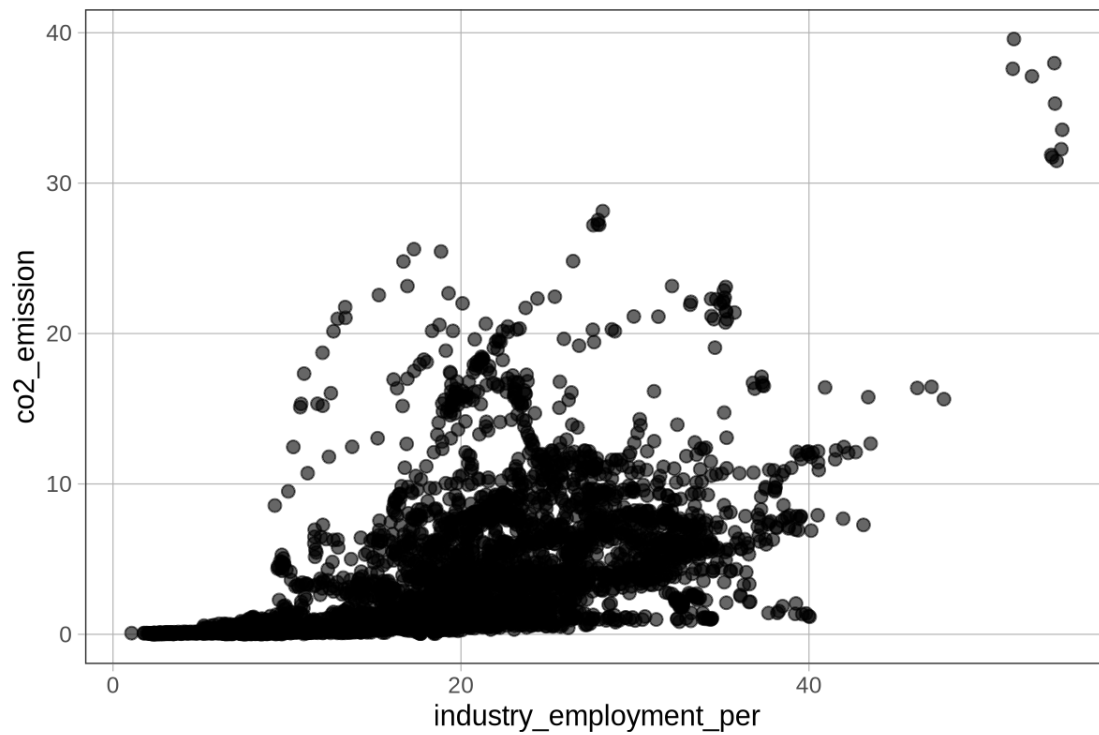
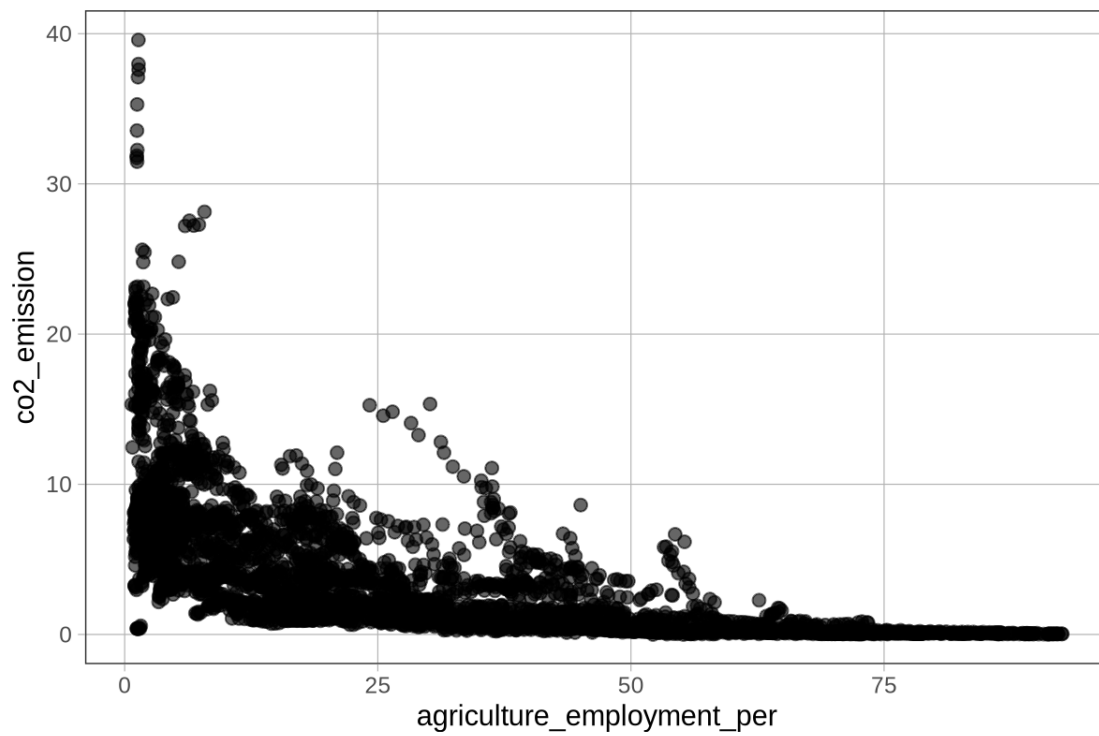


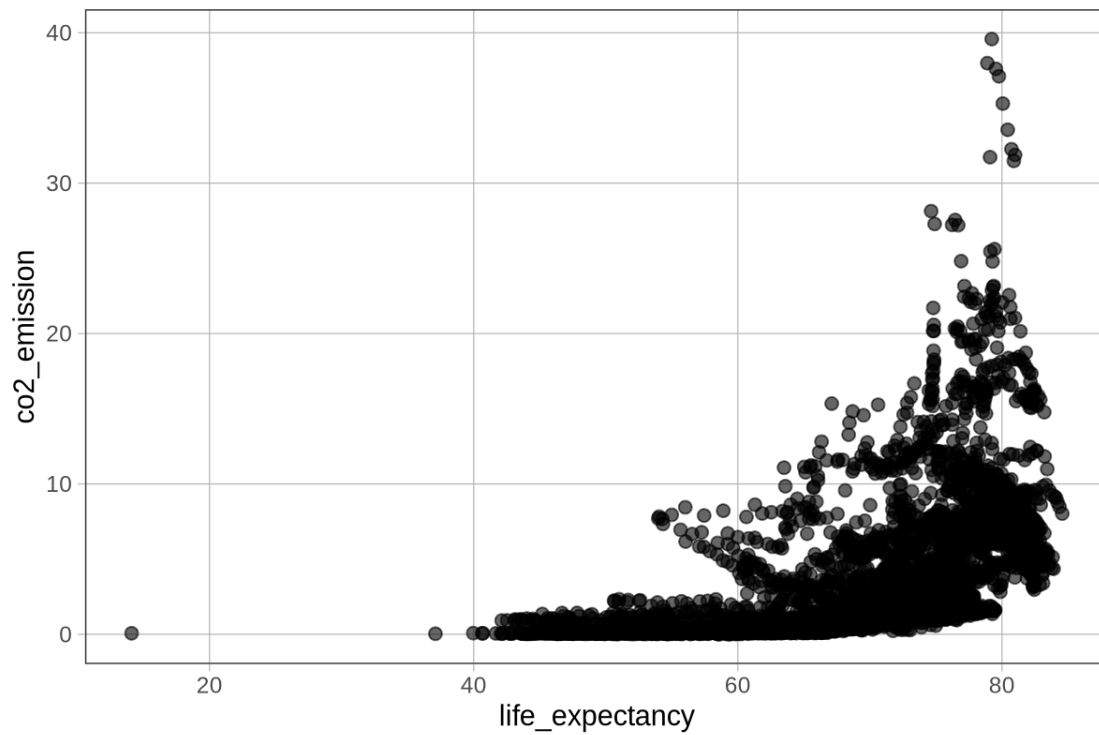
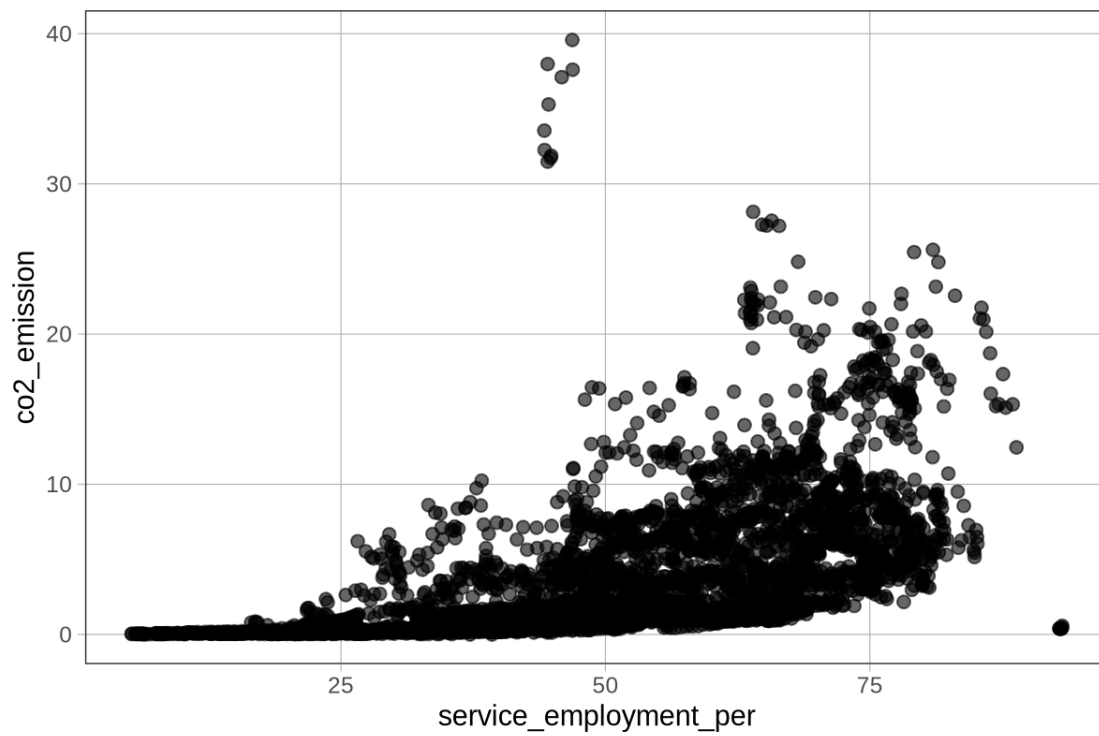


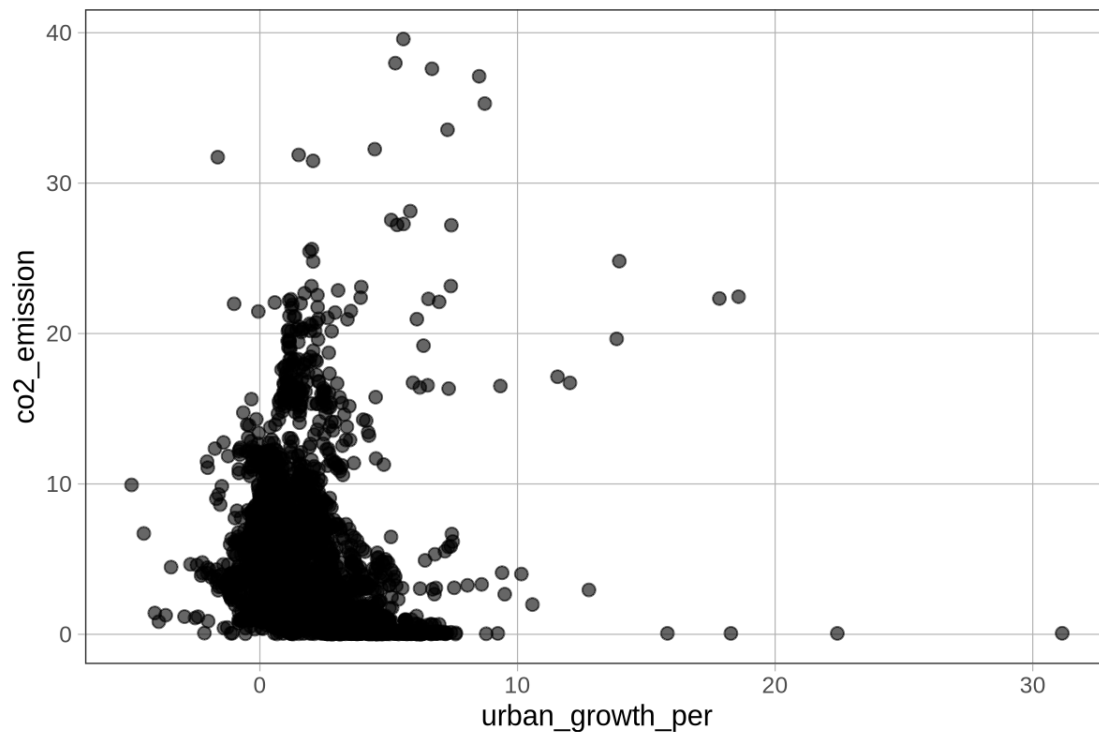
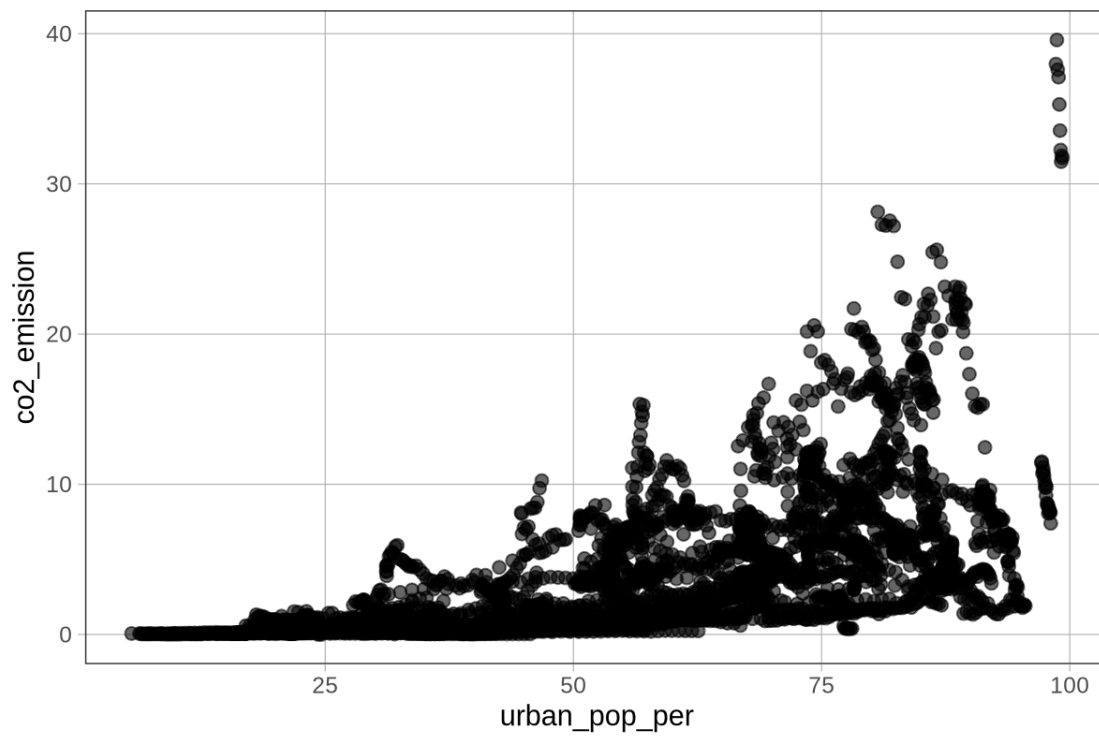


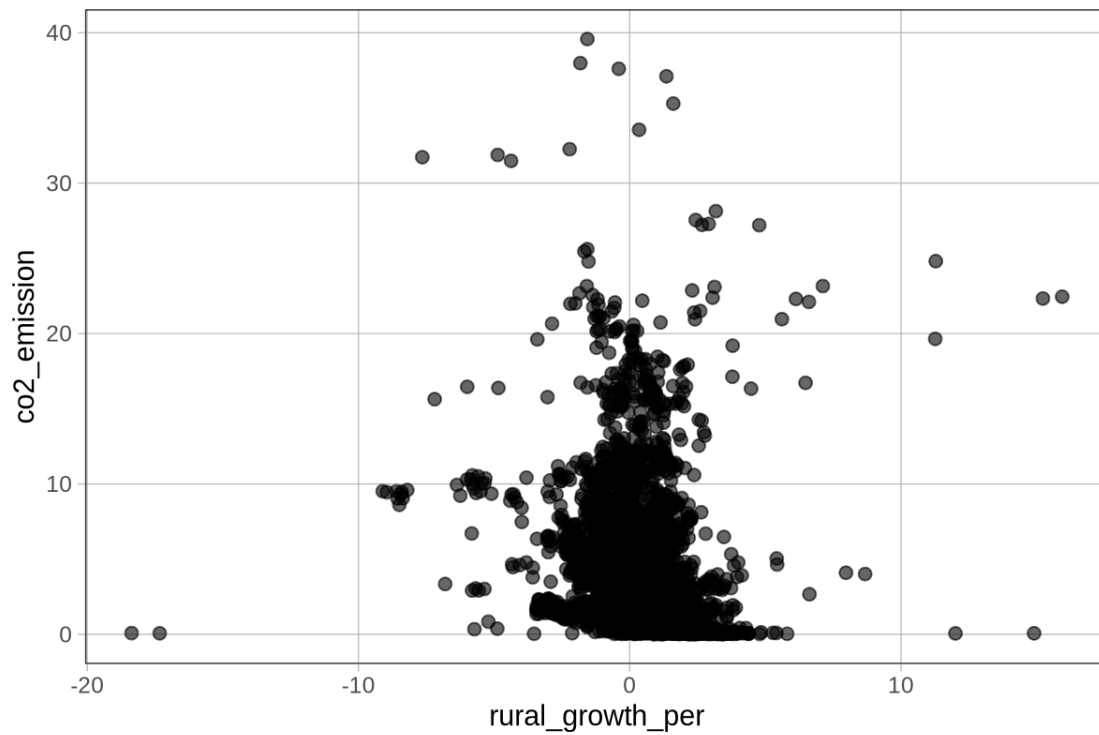
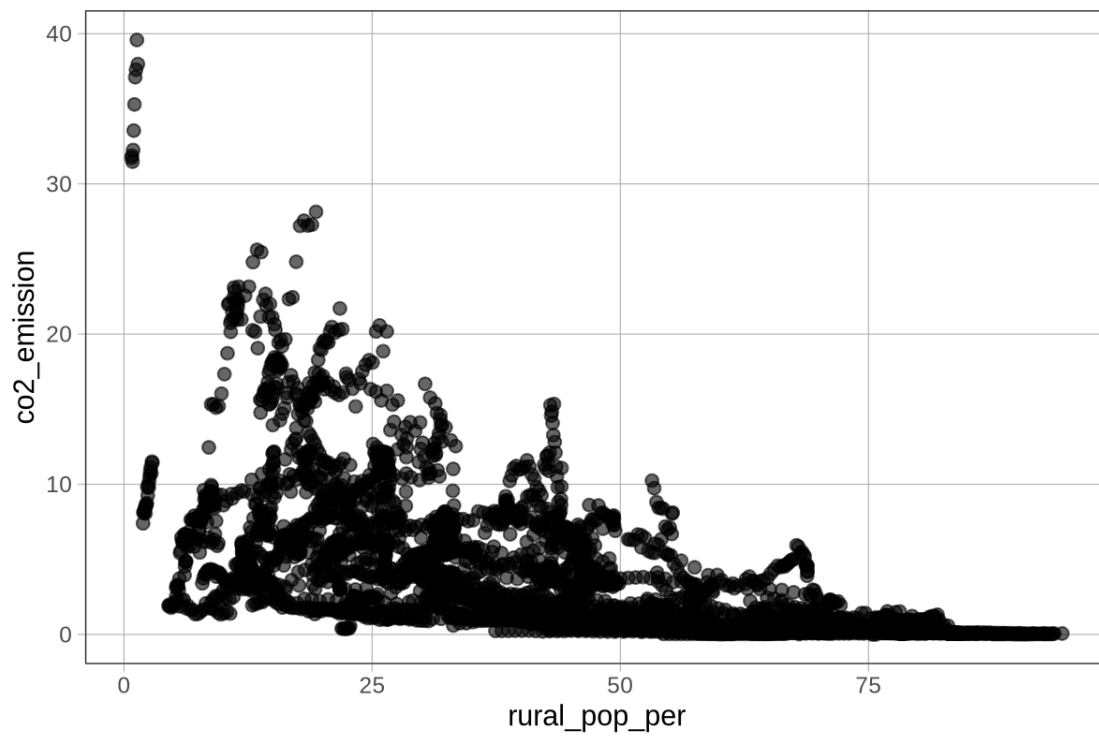












4102 different instances recorded, 24 different categories measured Y axis = co2 total x axis = year

country: We cant read

co2_emission: Most countries are less than 10ish

year: There are a bunch of really big spikes every 5ish years

ag_land_per: Mostly in the middle

area: Most countries are pretty small

population: Most countries are pretty small

gdp: Most countries are pretty small

ag_gdp_per: Mostly below 20, but not 0

ind_gdp_per: Usually around 25

man_gdp_per: Big spike around 10-15ish

serv_gdp_per: Big spike at roughly 55, but very nice triangle

imp_goods_cur: Mostly crammed on one edge

exp_goods_cur: Mostly crammed on one edge

imp_goods_gdp_per: Peaks around 30ish

exp_goods_gdp_per: Same as above, but slightly further left peak

tra_gdp_per: Again, pretty much the same

ag_emp_per: Most countries are at like 5ish, with a steady but spiky decline

ind_emp_per: Most countries are around 20ish

1.2 Useful variables:

- Agriculture stuff
- Urban/Rural population
- Service employment per capita
- Life expectancy

1.3 Part 2: Data Processing & Baseline Model

We will now prepare our data for modeling and fit a baseline model. Our baseline model's accuracy will be used as a comparison point for more advanced models. In this section, we will: - Split our data into a train and test set (code provided) - Fit a baseline model to predict **co2_emission**. This model will simply be the mean of **co2_emission** in the train data (i.e. the mean model) - Test the prediction accuracy of our baseline model by calculating its RMSE on the test set

Since we don't have to one-hot encode any of our data, we can go straight to splitting it into a train and test set. We'll do an 80-20 train/test split:

```
[5]: # Set seed so that we get the same random sample each time
set.seed(321)

# Get total number of rows (N) in dataset
N = nrow(co2_data)

# Get size of train set so that it's 80% of dataset, round to the nearest whole
  ↳ number
train_size = round(0.8 * N)

# Sample random row indices from the full dataset, without replacement
sample <- sample.int(n=N, size = train_size, replace = F)

# Select the sampled row indices for train set, remaining go to test set
train <- co2_data[sample,]
test  <- co2_data[-sample, ]

# Print the dimensions of our train and test sets
dim(train)
dim(test)
```

1. 3282 2. 25

1. 820 2. 25

```
[6]: mean_model <- mean(train$co2_emission)
mean_model

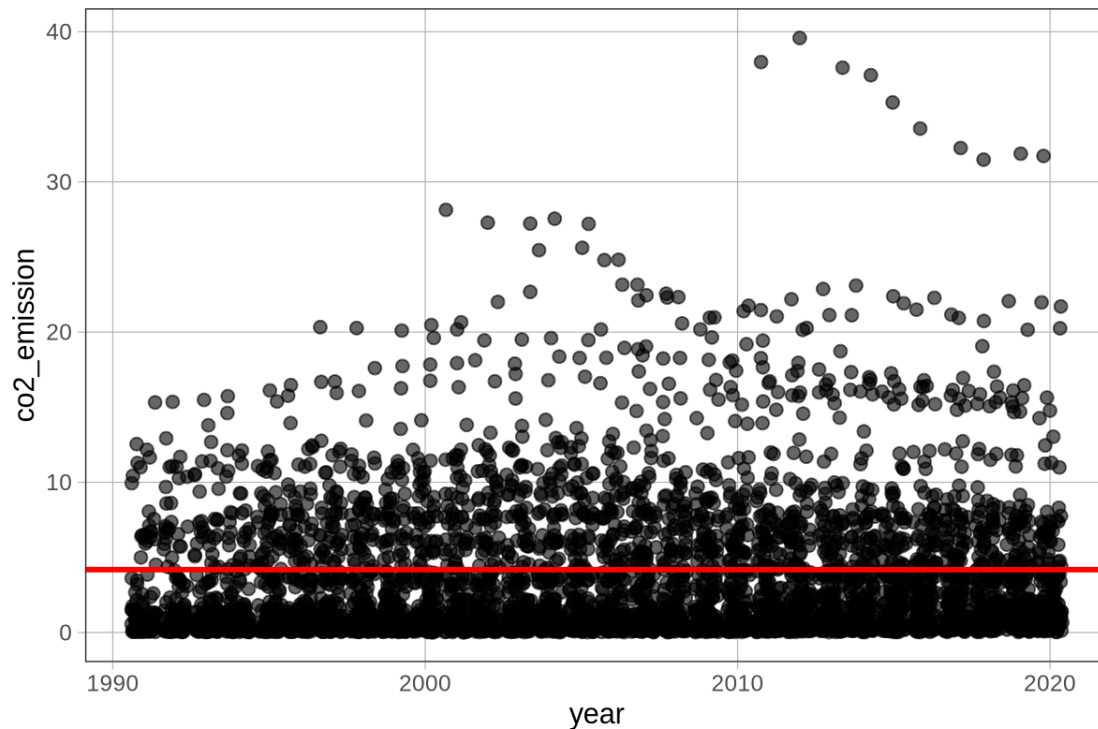
gf_jitter(co2_emission ~ year, data = co2_data) %>%
gf_hline(yintercept = mean_model, color= "red")

mean_model_error <- test$co2_emission - mean_model
sqrt(mean(mean_model_error^2))

#RMSE of 4.504 means that on average, the test data differs from the mean of
  ↳ the training data by 4.504 metric tons of co2 per capita
```

4.18513920035253

4.50449527654066



1.4 Part 3: Advanced Models - Regression

In this section, we will: - Train different multiple regression models on the train set - Test (on the test set) the prediction accuracy from our models, using RMSE - Interpret the coefficients from the most accurate multiple regression model to see which factors appear to be most important for predicting `co2_emission`

```
[7]: linear_model_check <- lm(co2_emission ~ year + agriculture_land_per + area +
  population + gdp + agriculture_gdp_per + industry_gdp_per +
  ↪+
  manufacturing_gdp_per + service_gdp_per + ↪
  ↪import_goods_currency +
  export_goods_currency + trade_gdp_per + ↪
  ↪agriculture_employment_per +
  industry_employment_per + service_employment_per + ↪
  ↪life_expectancy +
  urban_pop_per + urban_growth_per + rural_pop_per + ↪
  ↪rural_growth_per, data = train)
linear_model_check
summary(linear_model_check)
```

Call:

```
lm(formula = co2_emission ~ year + agriculture_land_per + area +
    population + gdp + agriculture_gdp_per + industry_gdp_per +
    manufacturing_gdp_per + service_gdp_per + import_goods_currency +
    export_goods_currency + trade_gdp_per + agriculture_employment_per +
    industry_employment_per + service_employment_per + life_expectancy +
    urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
    data = train)
```

Coefficients:

(Intercept)	year
4.184e+04	-5.111e-02
agriculture_land_per	area
3.614e-03	4.735e-07
population	gdp
-5.085e-09	-5.543e-14
agriculture_gdp_per	industry_gdp_per
5.680e-02	1.213e-01
manufacturing_gdp_per	service_gdp_per
-1.207e-01	4.862e-02
import_goods_currency	export_goods_currency
2.003e-12	2.442e-12
trade_gdp_per	agriculture_employment_per
1.577e-02	-4.176e+02
industry_employment_per	service_employment_per
-4.174e+02	-4.176e+02
life_expectancy	urban_pop_per
1.060e-01	5.439e-02
urban_growth_per	rural_pop_per
7.716e-02	NA
rural_growth_per	
2.030e-01	

Call:

```
lm(formula = co2_emission ~ year + agriculture_land_per + area +
    population + gdp + agriculture_gdp_per + industry_gdp_per +
    manufacturing_gdp_per + service_gdp_per + import_goods_currency +
    export_goods_currency + trade_gdp_per + agriculture_employment_per +
    industry_employment_per + service_employment_per + life_expectancy +
    urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
    data = train)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.7889	-1.9052	-0.2665	1.3243	23.2120

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.184e+04	3.360e+04	1.245	0.2131
year	-5.111e-02	7.236e-03	-7.064	1.97e-12 ***
agriculture_land_per	3.614e-03	2.844e-03	1.271	0.2038
area	4.735e-07	3.795e-08	12.477	< 2e-16 ***
population	-5.085e-09	5.402e-10	-9.413	< 2e-16 ***
gdp	-5.543e-14	1.455e-13	-0.381	0.7033
agriculture_gdp_per	5.680e-02	1.286e-02	4.418	1.03e-05 ***
industry_gdp_per	1.213e-01	9.976e-03	12.164	< 2e-16 ***
manufacturing_gdp_per	-1.207e-01	1.033e-02	-11.689	< 2e-16 ***
service_gdp_per	4.862e-02	1.104e-02	4.403	1.10e-05 ***
import_goods_currency	2.003e-12	1.910e-12	1.049	0.2943
export_goods_currency	2.442e-12	1.471e-12	1.660	0.0970 .
trade_gdp_per	1.577e-02	1.438e-03	10.968	< 2e-16 ***
agriculture_employment_per	-4.176e+02	3.360e+02	-1.243	0.2141
industry_employment_per	-4.174e+02	3.360e+02	-1.242	0.2142
service_employment_per	-4.176e+02	3.360e+02	-1.243	0.2141
life_expectancy	1.060e-01	1.144e-02	9.263	< 2e-16 ***
urban_pop_per	5.439e-02	4.735e-03	11.488	< 2e-16 ***
urban_growth_per	7.716e-02	3.889e-02	1.984	0.0474 *
rural_pop_per	NA	NA	NA	NA
rural_growth_per	2.030e-01	4.060e-02	5.000	6.04e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.109 on 3262 degrees of freedom

Multiple R-squared: 0.5873, Adjusted R-squared: 0.5849

F-statistic: 244.3 on 19 and 3262 DF, p-value: < 2.2e-16

```
linear_model_try1 <- lm(co2_emission ~ year + area + population + agriculture_gdp_per
+ industry_gdp_per + manufacturing_gdp_per + service_gdp_per + trade_gdp_per +
life_expectancy + urban_pop_per + rural_growth_per, data = train) linear_model_try1 sum-
mary(linear_model_try1) linear_model_try2 <- lm(co2_emission ~ year + agriculture_land_per
+ area + population + agriculture_gdp_per + industry_gdp_per + manufacturing_gdp_per
+ service_gdp_per + import_goods_currency + export_goods_currency + trade_gdp_per
+ agriculture_employment_per + industry_employment_per + service_employment_per +
life_expectancy + urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
data = train) linear_model_try2 summary(linear_model_try2)
```

1.5 INTERPRETATIONS OF COEFFICIENTS:

Area: For each 1 km² increase in area, the Co2 emissions are predicted to increase by 3.614e-03 metric tons per capita when controlling for all other variables.

industry_gdp_per: For each 1% increase in industry's contribution to the country's GDP per capita, the Co2 emissions are predicted to increase by 1.213e-01 metric tons per capita when controlling for all other variables.

manufacturing_gdp_per: For each 1% increase in manufacturing's contribution to the country's GDP per capita, the Co2 emissions are predicted to decrease by 1.207e-01 metric tons per capita when controlling for all other variables.

trade_gdp_per: For each 1% increase in trade's contribution to the country's GDP per capita, the Co2 emissions are predicted to increase by 1.577e-02 metric tons per capita when controlling for all other variables.

urban_pop_per: For each 1% increase of the percentage of the country's population that lives in an urban area, the Co2 emissions are predicted to increase by 7.716e-02 metric tons per capita when controlling for all other variables.

```
[8]: lmc_error <- test$co2_emission - predict(linear_model_check, newdata = test)
      sqrt(mean(lmc_error^2))
      #lmt1_error <- test$co2_emission - predict(linear_model_try1, newdata = test)
      #sqrt(mean(lmt1_error^2))
      #lmt2_error <- test$co2_emission - predict(linear_model_try2, newdata = test)
      #sqrt(mean(lmt2_error^2))
```

Warning message in predict.lm(linear_model_check, newdata = test):
 "prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases"

2.8775800017728

```
complex_model_check <- lm(co2_emission ~ poly(industry_gdp_per, 6) +
  poly(manufacturing_gdp_per, 6) + poly(area, 6) + poly(trade_gdp_per, 6) + poly(population,
  6) + poly(urban_pop_per, 6), data = train) cmc_error <- test$co2_emission -
  predict(complex_model_check, newdata = test) sqrt(mean(cmc_error^2)) complex_model_check2 <-
  lm(co2_emission ~ poly(industry_gdp_per, 6) + poly(manufacturing_gdp_per, 6) + poly(area, 6) +
  poly(trade_gdp_per, 6) + poly(urban_pop_per, 6), data = train) cmc_error2 <- test$co2_emission -
  predict(complex_model_check2, newdata = test) sqrt(mean(cmc_error2^2)) complex_model_check3
<- lm(co2_emission ~ poly(industry_gdp_per, 6) + poly(manufacturing_gdp_per, 6) +
  poly(area, 6) + poly(trade_gdp_per, 6) + poly(population, 6) + poly(life_expectancy,
  6) + poly(urban_pop_per, 6), data = train) cmc_error3 <- test$co2_emission -
  predict(complex_model_check3, newdata = test) sqrt(mean(cmc_error3^2))
```

```
[9]: complex_model_check4 <- lm(co2_emission ~ poly(year, 7) +
  poly(area, 7) +
  poly(gdp, 7) +
  poly(industry_gdp_per, 7) +
  poly(service_gdp_per, 7) +
  poly(export_goods_currency, 7) +
  poly(agriculture_employment_per, 7) +
  poly(service_employment_per, 7) +
  poly(agriculture_land_per, 7) +
  poly(population, 7) +
  poly(manufacturing_gdp_per, 7) +
  poly(import_goods_currency, 7) +
  poly(trade_gdp_per, 7) +
  poly(industry_employment_per, 7) +
```

```

poly(life_expectancy, 7) +
poly(urban_pop_per, 7) +
poly(urban_growth_per, 7) +
poly(rural_growth_per, 7), data = train)
cmc_error4 <- test$co2_emission - predict(complex_model_check4, newdata = test)
sqrt(mean(cmc_error4^2))

```

1.97607006912232

1.6 Part 4: Advanced Models - Neural Networks

In this section, we will... - Make all quantitative predictors vary between 0 - 1 (code provided) - Fit several neural network models on the train data - Test (on the test set) the prediction accuracy of our models using RMSE - Compare the RMSE of our best neural network with the RMSE of our best regression model

Let's download our neural network package:

```

[10]: # Load neuralnet package (R's main neural network package)
install.packages('neuralnet')
library(neuralnet)

```

Updating HTML index of packages in '.Library'

Making 'packages.html' ...
done

Attaching package: 'neuralnet'

The following object is masked from 'package:dplyr':

compute

Now, let's make all our quantitative predictors vary between 0 - 1 by dividing each by their maxima. Note that we are storing these scaled values in **new dataframes called train_net and test_net**. We'll use the new dataframes for training and testing our neural networks.

```

[11]: # Rescale values in train set, store in new dataframe called train_net
train_net <- train

train_net$industry_gdp_per <- train$industry_gdp_per/
  ↪max(co2_data$industry_gdp_per)
train_net$life_expectancy <- train$life_expectancy/max(co2_data$life_expectancy)
train_net$industry_employment_per <- train$industry_employment_per/
  ↪max(co2_data$industry_employment_per)

```

```

train_net$export_goods_gdp_per <- train$export_goods_gdp_per/
  ↪max(co2_data$export_goods_gdp_per)
train_net$manufacturing_gdp_per <- train$manufacturing_gdp_per/
  ↪max(co2_data$manufacturing_gdp_per)
train_net$service_gdp_per <- train$service_gdp_per/max(co2_data$service_gdp_per)
train_net$area <- train$area/max(co2_data$area)
train_net$trade_gdp_per <- train$trade_gdp_per/max(co2_data$trade_gdp_per)
train_net$gdp <- train$gdp/max(co2_data$gdp)
train_net$population <- train$population/max(co2_data$population)
train_net$urban_pop_per <- train$urban_pop_per/max(co2_data$urban_pop_per)
train_net$export_goods_currency <- train$export_goods_currency/
  ↪max(co2_data$export_goods_currency)
train_net$agriculture_land_per <- train$agriculture_land_per/
  ↪max(co2_data$agriculture_land_per)
train_net$agriculture_gdp_per <- train$agriculture_gdp_per/
  ↪max(co2_data$agriculture_gdp_per)
train_net$import_goods_currency <- train$import_goods_currency/
  ↪max(co2_data$import_goods_currency)
train_net$import_goods_gdp_per <- train$import_goods_gdp_per/
  ↪max(co2_data$import_goods_gdp_per)
train_net$agriculture_employment_per <- train$agriculture_employment_per/
  ↪max(co2_data$agriculture_employment_per)
train_net$service_employment_per <- train$service_employment_per/
  ↪max(co2_data$service_employment_per)
train_net$urban_growth_per <- train$urban_growth_per/
  ↪max(co2_data$urban_growth_per)
train_net$rural_pop_per <- train$rural_pop_per/max(co2_data$rural_pop_per)
train_net$rural_growth_per <- train$rural_growth_per/
  ↪max(co2_data$rural_pop_perrural_growth_per)

# Rescale values in test set, store in new dataframe called test_net
test_net <- test

test_net$industry_gdp_per <- test$industry_gdp_per/
  ↪max(co2_data$industry_gdp_per)
test_net$life_expectancy <- test$life_expectancy/max(co2_data$life_expectancy)
test_net$industry_employment_per <- test$industry_employment_per/
  ↪max(co2_data$industry_employment_per)
test_net$export_goods_gdp_per <- test$export_goods_gdp_per/
  ↪max(co2_data$export_goods_gdp_per)
test_net$manufacturing_gdp_per <- test$manufacturing_gdp_per/
  ↪max(co2_data$manufacturing_gdp_per)
test_net$service_gdp_per <- test$service_gdp_per/max(co2_data$service_gdp_per)
test_net$area <- test$area/max(co2_data$area)
test_net$trade_gdp_per <- test$trade_gdp_per/max(co2_data$trade_gdp_per)

```

```

test_net$gdp <- test$gdp/max(co2_data$gdp)
test_net$population <- test$population/max(co2_data$population)
test_net$urban_pop_per <- test$urban_pop_per/max(co2_data$urban_pop_per)
test_net$export_goods_currency <- test$export_goods_currency/
  ↪max(co2_data$export_goods_currency)
test_net$agriculture_land_per <- test$agriculture_land_per/
  ↪max(co2_data$agriculture_land_per)
test_net$agriculture_gdp_per <- test$agriculture_gdp_per/
  ↪max(co2_data$agriculture_gdp_per)
test_net$import_goods_currency <- test$import_goods_currency/
  ↪max(co2_data$import_goods_currency)
test_net$import_goods_gdp_per <- test$import_goods_gdp_per/
  ↪max(co2_data$import_goods_gdp_per)
test_net$agriculture_employment_per <- test$agriculture_employment_per/
  ↪max(co2_data$agriculture_employment_per)
test_net$service_employment_per <- test$service_employment_per/
  ↪max(co2_data$service_employment_per)
test_net$urban_growth_per <- test$urban_growth_per/
  ↪max(co2_data$urban_growth_per)
test_net$rural_pop_per <- test$rural_pop_per/max(co2_data$rural_pop_per)
test_net$rural_growth_per <- test$rural_growth_per/
  ↪max(co2_data$rural_growth_per)

head(train_net)
head(test_net)

```

Warning message in base::max(x, ..., na.rm = na.rm):
 "no non-missing arguments to max; returning -Inf"

		country	year	area	agriculture_land_per	population	gdp
		<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
A data.frame: 6 × 25	4082	Zimbabwe	1996	0.023615212	0.4304403	0.007921601	0.000400
	333	Belgium	2002	0.001848439	0.5366309	0.007322504	0.012084
	1017	Cuba	1999	0.006556220	0.7253840	0.007845010	0.001326
	2618	Morocco	2010	0.027244330	0.7860008	0.023006778	0.004717
	3710	Togo	2013	0.003320231	0.8215658	0.005035950	0.000274
	1071	Czechia	1999	0.004716938	0.6482369	0.007287832	0.003048
		country	year	area	agriculture_land_per	population	gdp
		<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
A data.frame: 6 × 25	3	Albania	1998	0.00167263	0.4862630	0.002217086	0.0001190760
	5	Albania	2000	0.00167263	0.4883976	0.002189091	0.0001627779
	8	Albania	2003	0.00167263	0.4785784	0.002154076	0.0002624525
	10	Albania	2005	0.00167263	0.4597939	0.002134141	0.0003765997
	16	Albania	2011	0.00167263	0.5127321	0.002058816	0.0006029073
	24	Albania	2019	0.00167263	0.5012052	0.002022671	0.0007203511

Important note to participants: When using the `neuralnet` command, we recommend setting the argument `threshold` to a value somewhere between `threshold = 10` and `threshold = 20`, to give your model the highest chance of properly fitting.

1.7 Neural Network Try 1

```
set.seed(100) network_try1 <- neuralnet(co2_emission~ agriculture_land_per + area + pop-
ulation + gdp + agriculture_gdp_per + industry_gdp_per + manufacturing_gdp_per +
service_gdp_per + import_goods_currency + export_goods_currency + trade_gdp_per
+ agriculture_employment_per + industry_employment_per + service_employment_per +
life_expectancy + urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
data = train_net, lifesign = 'full', threshold = 10, hidden=c(4, 2)) nt1_pred <- pre-
dict(network_try1, newdata = test_net) nt1_error <- test_net$co2_emission - nt1_pred
sqrt(mean(nt1_error^2))
```

1.8 Neural Network Try 2

```
set.seed(100) network_try2 <- neuralnet(co2_emission~ agriculture_land_per + area + pop-
ulation + gdp + agriculture_gdp_per + industry_gdp_per + manufacturing_gdp_per +
service_gdp_per + import_goods_currency + export_goods_currency + trade_gdp_per
+ agriculture_employment_per + industry_employment_per + service_employment_per +
life_expectancy + urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
data = train_net, lifesign = 'full', threshold = 10, hidden=c(16,8,4,2)) nt2_pred <- pre-
dict(network_try2, newdata = test_net) nt2_error <- test_net$co2_emission - nt2_pred
sqrt(mean(nt2_error^2))
```

1.9 Neural Network Try 4

```
set.seed(100) network_try4 <- neuralnet(co2_emission~ agriculture_land_per + area + pop-
ulation + gdp + agriculture_gdp_per + industry_gdp_per + manufacturing_gdp_per +
service_gdp_per + import_goods_currency + export_goods_currency + trade_gdp_per
+ agriculture_employment_per + industry_employment_per + service_employment_per +
life_expectancy + urban_pop_per + urban_growth_per + rural_pop_per + rural_growth_per,
data = train_net, lifesign = 'full', threshold = 5, hidden=c(8,4,2)) nt4_pred <- pre-
dict(network_try4, newdata = test_net) nt4_error <- test_net$co2_emission - nt4_pred
sqrt(mean(nt4_error^2))
```

```
[12]: set.seed(100)
network_try3 <- neuralnet(co2_emission~ agriculture_land_per + area +
                           population + gdp + agriculture_gdp_per +
                           ↪industry_gdp_per +
                           manufacturing_gdp_per + service_gdp_per +
                           ↪import_goods_currency +
                           export_goods_currency + trade_gdp_per +
                           ↪agriculture_employment_per +
                           industry_employment_per + service_employment_per +
                           ↪life_expectancy + urban_pop_per +
```

```

        urban_growth_per + rural_pop_per + rural_growth_per,
data = train_net, lifestage = 'full', threshold = 10, hidden=c(36))
nt3_pred <- predict(network_try3, newdata = test_net)
nt3_error <- test_net$co2_emission - nt3_pred
sqrt(mean(nt3_error^2))

```

```

hidden: 36    thresh: 10    rep: 1/1    steps:
1000 min thresh: 28.6410539673848

2000 min thresh: 17.4467960828093

3000 min thresh: 10.829234321701

3384
error: 722.91227
time: 31.56 secs

```

0.780058401586964

The RMSE of our best neural network had an RMSE of around **0.780058401586964**, which is significantly better than our best regression model that had an RMSE of **1.97607006912232**.

1.10 Part 5: Evaluation & Conclusion

In this section, we will... - Print the **actual** number of metric tons of CO2 emission per capita in 2020 (the latest year in our dataset), arranging the countries by order of emission rates (code provided). - Use our best model to **predict** the number of metric tons of CO2 emission per capita for each country in 2020 (code partially provided). Then, we'll compare our model's 2020 CO2 emission predictions to the actual 2020 CO2 emission values. This will allow us to evaluate (using the latest available data in our dataset) whether our model is reasonably reliable for predicting CO2 emissions. - Answer these key questions: What key predictors should policymakers and government officials be looking at when drafting CO2 targets? What predictors should countries focus on if they'd like to reduce their overall emissions? We'll answer these questions using the coefficients from our best regression model. - Discuss limitations and further work that could be done to improve our model.

1.11 Part 5 Answers

- Government officials should be looking at factors such as the percentage of a country's GDP that comes from industry, trade, and manufacturing. In addition they should look at the percentage of the population that lives in an urban environment. While life expectancy is a great predictor of overall carbon emissions it shouldn't be considered as a practical way to decrease carbon emissions because there is little proof that life expectancy directly decreases carbon emissions. Overall policy makers should consider as many factors as possible in order to see the maximum amount of impact.
- One of the ways we could make our model better is adding factors that aren't considered in the current dataset. This could be like the amount of animal agriculture in a country. Animal

agriculture accounts for a large part of CO2 emissions and it isn't currently recognized by our model. We see this inaccuracy in effect when seeing the predicted top 5 emitters of CO2 and the actual top 5 emitters. Furthermore this means that there are unforeseen factors and other pieces of data that we need to add to our model to make it more accurate.

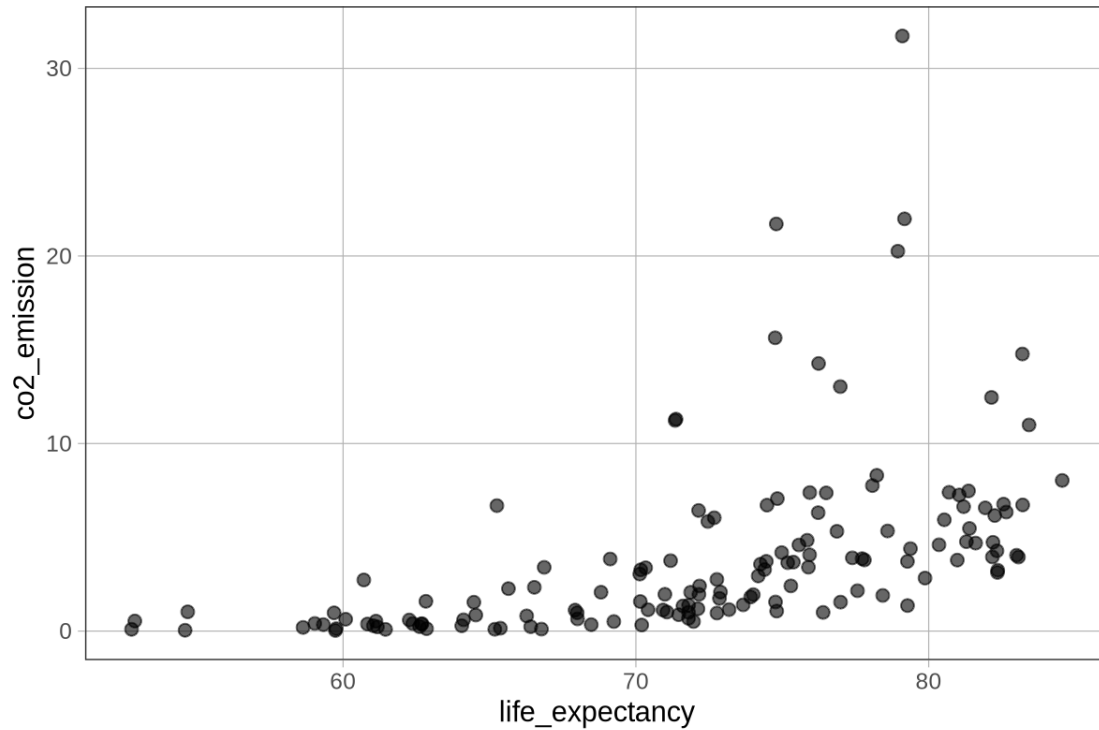
First, let's filter our dataset for CO2 emission in 2020 and print out our results (countries arranged by the biggest CO2 emission contributors)

```
[13]: # Filter for the CO2 emission in 2020
year_2020 <- filter(co2_data, year==2020)

# Arrange by highest CO2 emissions in 2020
arrange(year_2020, desc(co2_emission))

gf_jitter(co2_emission ~ life_expectancy, data = year_2020)
```

	country <chr>	year <int>	area <dbl>	agriculture_land_per <dbl>	population <int>
	Qatar	2020	11490.00	6.440383	2760385
	Bahrain	2020	785.00	10.318471	1477469
	Brunei Darussalam	2020	5270.00	2.542694	441725
	United Arab Emirates	2020	71020.00	5.488595	9287289
	Oman	2020	309500.00	4.713829	4543399
	Australia	2020	7692020.00	46.252480	25655289
	Saudi Arabia	2020	2149690.00	80.773214	35997107
	United States	2020	9147420.00	44.363367	331511512
	Luxembourg	2020	2574.46	51.325715	630419
	Kazakhstan	2020	2699700.00	79.269252	18755666
	Russian Federation	2020	16376870.00	13.158436	144073139
	Korea, Rep.	2020	97600.00	16.608607	51836239
	Czechia	2020	77198.53	45.646867	10697858
	Japan	2020	364500.00	12.844993	126261000
	China	2020	9388210.00	55.462649	1411100000
	Netherlands	2020	33670.00	53.889219	17441500
	Belgium	2020	30280.00	45.066357	11538604
	Malaysia	2020	328550.00	26.087354	33199993
	Poland	2020	306130.00	48.126613	37899070
	Germany	2020	349390.00	47.497066	83160871
	Iran, Islamic Rep.	2020	1622500.00	28.984694	87290193
	Ireland	2020	68890.00	65.495718	4985382
	Norway	2020	364285.00	2.706568	5379475
	Serbia	2020	84090.00	41.408015	6899126
	South Africa	2020	1213090.00	79.417850	58801927
	Austria	2020	82520.00	31.539813	8916864
	Finland	2020	303940.00	7.468579	5529543
	Mongolia	2020	1557506.80	72.358944	3294335
	Israel	2020	21640.00	29.870610	9215100
A data.frame: 144 × 25	Bosnia and Herzegovina	2020	51200.00	43.789062	3318407
	Benin	2020	112760	35.030153	12643123
	Ghana	2020	227533	55.393020	32180401
	Angola	2020	1246700	36.810780	33428486
	Nigeria	2020	910770	75.203399	208327405
	Zimbabwe	2020	386850	41.876696	15669666
	Bangladesh	2020	130170	76.269494	167420951
	Nepal	2020	143350	28.747820	29348627
	Cote d'Ivoire	2020	318000	73.270440	26811790
	Zambia	2020	743390	32.067959	18927715
	Djibouti	2020	23180	73.507334	1090156
	Cameroon	2020	472710	20.625754	26491087
	Kenya	2020	569140	48.687493	51985780
	Guinea	2020	245720	59.559661	13205153
	Timor-Leste	2020	14870	22.958978	1299995
	Solomon Islands	2020	27990	4.287245	691191
	Togo	2020	54390	70.233499	8442580
	Haiti	2020	27560	64.949202	11306801
	Gambia, The	2020	10120	62.648221	2573995
	Tanzania	2020	885800	44.616392	61704518
	Mozambique	2020	786380	52.663893	31178239



We will now use our best model to predict the CO2 emission per country for 2020. We'll add our predictions as a column in our dataset and arrange the data based on that column to get our predicted CO2 emission contributions.

```
[14]: ## If your best model IS a neural network, use this code block

best_model <- network_try3

# Scale the 2020 dataset
year_2020_net <- year_2020
year_2020_net$industry_gdp_per <- year_2020$industry_gdp_per /
  ↪max(co2_data$industry_gdp_per)
year_2020_net$life_expectancy <- year_2020$life_expectancy /
  ↪max(co2_data$life_expectancy)
year_2020_net$industry_employment_per <- year_2020$industry_employment_per /
  ↪max(co2_data$industry_employment_per)
year_2020_net$export_goods_gdp_per <- year_2020$export_goods_gdp_per /
  ↪max(co2_data$export_goods_gdp_per)
year_2020_net$manufacturing_gdp_per <- year_2020$manufacturing_gdp_per /
  ↪max(co2_data$manufacturing_gdp_per)
year_2020_net$service_gdp_per <- year_2020$service_gdp_per /
  ↪max(co2_data$service_gdp_per)
year_2020_net$area <- year_2020$area / max(co2_data$area)
```

```

year_2020_net$trade_gdp_per <- year_2020$trade_gdp_per/
  ↪max(co2_data$trade_gdp_per)
year_2020_net$gdp <- year_2020$gdp/max(co2_data$gdp)
year_2020_net$population <- year_2020$population/max(co2_data$population)
year_2020_net$urban_pop_per <- year_2020$urban_pop_per/
  ↪max(co2_data$urban_pop_per)
year_2020_net$export_goods_currency <- year_2020$export_goods_currency/
  ↪max(co2_data$export_goods_currency)
year_2020_net$agriculture_land_per <- year_2020$agriculture_land_per/
  ↪max(co2_data$agriculture_land_per)
year_2020_net$agriculture_gdp_per <- year_2020$agriculture_gdp_per/
  ↪max(co2_data$agriculture_gdp_per)
year_2020_net$import_goods_currency <- year_2020$import_goods_currency/
  ↪max(co2_data$import_goods_currency)
year_2020_net$import_goods_gdp_per <- year_2020$import_goods_gdp_per/
  ↪max(co2_data$import_goods_gdp_per)
year_2020_net$agriculture_employment_per <-
  ↪year_2020$agriculture_employment_per/max(co2_data$agriculture_employment_per)
year_2020_net$service_employment_per <- year_2020$service_employment_per/
  ↪max(co2_data$service_employment_per)
year_2020_net$urban_growth_per <- year_2020$urban_growth_per/
  ↪max(co2_data$urban_growth_per)
year_2020_net$rural_pop_per <- year_2020$rural_pop_per/
  ↪max(co2_data$rural_pop_per)
year_2020_net$rural_growth_per <- year_2020$rural_growth_per/
  ↪max(co2_data$rural_growth_per)

# Predict CO2 emission for each country in 2020
predicted_co2_emission <- predict(best_model, newdata= year_2020_net)

# Add predictions to colum
year_2020$pred_co2_emission <- predicted_co2_emission

# Display dataset, arranged in desc order
arrange(year_2020, desc(co2_emission))

```

	country <chr>	year <int>	area <dbl>	agriculture_land_per <dbl>	population <int>
	Qatar	2020	11490.00	6.440383	2760385
	Bahrain	2020	785.00	10.318471	1477469
	Brunei Darussalam	2020	5270.00	2.542694	441725
	United Arab Emirates	2020	71020.00	5.488595	9287289
	Oman	2020	309500.00	4.713829	4543399
	Australia	2020	7692020.00	46.252480	25655289
	Saudi Arabia	2020	2149690.00	80.773214	35997107
	United States	2020	9147420.00	44.363367	331511512
	Luxembourg	2020	2574.46	51.325715	630419
	Kazakhstan	2020	2699700.00	79.269252	18755666
	Russian Federation	2020	16376870.00	13.158436	144073139
	Korea, Rep.	2020	97600.00	16.608607	51836239
	Czechia	2020	77198.53	45.646867	10697858
	Japan	2020	364500.00	12.844993	126261000
	China	2020	9388210.00	55.462649	1411100000
	Netherlands	2020	33670.00	53.889219	17441500
	Belgium	2020	30280.00	45.066357	11538604
	Malaysia	2020	328550.00	26.087354	33199993
	Poland	2020	306130.00	48.126613	37899070
	Germany	2020	349390.00	47.497066	83160871
	Iran, Islamic Rep.	2020	1622500.00	28.984694	87290193
	Ireland	2020	68890.00	65.495718	4985382
	Norway	2020	364285.00	2.706568	5379475
	Serbia	2020	84090.00	41.408015	6899126
	South Africa	2020	1213090.00	79.417850	58801927
	Austria	2020	82520.00	31.539813	8916864
	Finland	2020	303940.00	7.468579	5529543
	Mongolia	2020	1557506.80	72.358944	3294335
	Israel	2020	21640.00	29.870610	9215100
A data.frame: 144 × 26	Bosnia and Herzegovina	2020	51200.00	43.789062	3318407
	Benin	2020	112760	35.030153	12643123
	Ghana	2020	227533	55.393020	32180401
	Angola	2020	1246700	36.810780	33428486
	Nigeria	2020	910770	75.203399	208327405
	Zimbabwe	2020	386850	41.876696	15669666
	Bangladesh	2020	130170	76.269494	167420951
	Nepal	2020	143350	28.747820	29348627
	Cote d'Ivoire	2020	318000	73.270440	26811790
	Zambia	2020	743390	32.067959	18927715
	Djibouti	2020	23180	73.507334	1090156
	Cameroon	2020	472710	20.625754	26491087
	Kenya	2020	569140	48.687493	51985780
	Guinea	2020	245720	59.559661	13205153
	Timor-Leste	2020	14870	22.958978	1299995
	Solomon Islands	2020	27990	4.287245	691191
	Togo	2020	54390	70.233499	8442580
	Haiti	2020	27560	64.949202	11306801
	Gambia, The	2020	10120	62.648221	2573995
	Tanzania	2020	885800	44.616392	61704518
	Mozambique	2020	786380	52.663893	31178239

```
[15]: testErrors <- year_2020$co2_emission - year_2020$pred_co2_emission  
testErrorsRMSE <- sqrt(mean(testErrors^2))  
testErrorsRMSE  
year_2020$testErrors <- testErrors  
arrange(year_2020, desc(pred_co2_emission))
```

1.0744624689467

	country <chr>	year <int>	area <dbl>	agriculture_land_per <dbl>	population <int>
	Qatar	2020	11490.00	6.440383	2760385
	Bahrain	2020	785.00	10.318471	1477469
	United Arab Emirates	2020	71020.00	5.488595	9287289
	Oman	2020	309500.00	4.713829	4543399
	Brunei Darussalam	2020	5270.00	2.542694	441725
	Luxembourg	2020	2574.46	51.325715	630419
	Saudi Arabia	2020	2149690.00	80.773214	35997107
	Australia	2020	7692020.00	46.252480	25655289
	United States	2020	9147420.00	44.363367	331511512
	Korea, Rep.	2020	97600.00	16.608607	51836239
	Kazakhstan	2020	2699700.00	79.269252	18755666
	Russian Federation	2020	16376870.00	13.158436	144073139
	Estonia	2020	42750.00	23.040936	1329522
	Japan	2020	364500.00	12.844993	126261000
	Netherlands	2020	33670.00	53.889219	17441500
	Czechia	2020	77198.53	45.646867	10697858
	Germany	2020	349390.00	47.497066	83160871
	Finland	2020	303940.00	7.468579	5529543
	Poland	2020	306130.00	48.126613	37899070
	Belgium	2020	30280.00	45.066357	11538604
	Slovenia	2020	20136.40	30.317733	2102419
	China	2020	9388210.00	55.462649	1411100000
	Mongolia	2020	1557506.80	72.358944	3294335
	New Zealand	2020	263310.00	38.562911	5090200
	Norway	2020	364285.00	2.706568	5379475
	Iran, Islamic Rep.	2020	1622500.00	28.984694	87290193
	Belarus	2020	202980.00	40.797123	9379952
	Cyprus	2020	9240.00	13.814859	1237537
	Ireland	2020	68890.00	65.495718	4985382
A data.frame: 144 × 27	South Africa	2020	1213090.00	79.417850	58801927
	Pakistan	2020	770880	47.637765	227196741
	Mauritania	2020	1030700	38.527215	4498604
	Senegal	2020	192530	48.979380	16436120
	Samoa	2020	2780	17.773381	214929
	Solomon Islands	2020	27990	4.287245	691191
	Guinea	2020	245720	59.559661	13205153
	Cambodia	2020	176520	34.247608	16396860
	Gambia, The	2020	10120	62.648221	2573995
	Timor-Leste	2020	14870	22.958978	1299995
	Benin	2020	112760	35.030153	12643123
	Togo	2020	54390	70.233499	8442580
	Angola	2020	1246700	36.810780	33428486
	Nepal	2020	143350	28.747820	29348627
	Ethiopia	2020	1128571	34.092752	117190911
	Cote d'Ivoire	2020	318000	73.270440	26811790
	Bangladesh	2020	130170	76.269494	167420951
	Mozambique	2020	786380	52.663893	31178239
	Niger	2020	1266700	36.784558	24333639
	Rwanda	2020	24670	79.391974	13146362
	Congo, Dem. Rep.	2020	2267050	14.951148	92853164